A REALIZATION OF G_2 FOR LITT PROBLEM NO. 13

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Abstract. A family of complex abelian threefolds is constructed from cyclic triple covers

$$C: \quad w^3 = y - c, \qquad c \neq 0,$$

of elliptic curves $y^2 = x^3 + Ax + B$. The induced Prym polarization has type $(1, 1, 3)$. The centred theta section X is a normal symmetric surface with a unique simple elliptic singularity of type $\widetilde{E}_7$, and $\chi(\mathrm{IC}_X) = 7$. For general parameters, the pair-incidence curve $X \cap (-x - X)$ has exactly two nonexceptional components. Their top cohomology gives an unshifted summand $\mathrm{IC}_X \subset \mathrm{Alt}^2(\mathrm{IC}_X)$ in the convolution category, while a generic unitary twist has Hodge multiplicities $(2, 3, 2)$. These facts determine the full Tannaka group as G_2 on its standard seven-dimensional representation for general parameters. A simultaneous resolution and nearby cycles extend this conclusion to every parameter for which the elliptic curve and branch divisor are smooth. In particular, the construction gives an explicit example defined over $\mathbf{Q}$ and realizes the G_2 case of Litt Problem No. 13.

1. Introduction

Let A be a complex abelian variety. After negligible objects are quotiented out, convolution turns the relevant constructible complexes on A into a neutral Tannakian category. Thus an integral subvariety $Z \subset A$ with $\chi(\mathrm{IC}_Z) \neq 0$ determines a Tannaka group $G(\mathrm{IC}_Z)$ and a faithful module of dimension $\chi(\mathrm{IC}_Z)$; see [9, Proposition 3.1 and Definition 3.2]. Litt Problem No. 13 asks whether the four exceptional simple types

$$G_2, \qquad E_7, \qquad E_8, \qquad F_4$$

can occur in this manner [15]. Only the G_2 case is considered here. The construction uses a singular theta surface, so the exclusion results for smooth subvarieties in [12] do not apply.

The construction is defined over a rational parameter space. Put

$$\Delta_E = 4A^3 + 27B^2, \qquad \Delta_{\mathrm{br}} = 4A^3 + 27(B - c^2)^2,$$

and let

$$\mathcal{B}_{\mathbf{Q}} = \operatorname{Spec} \mathbf{Q}[A, B, c, c^{-1}, \Delta_E^{-1}, \Delta_{\mathrm{br}}^{-1}], \qquad \mathcal{B} = \mathcal{B}_{\mathbf{Q}} \times_{\mathbf{Q}} \mathbf{C}.$$

For $b = (A, B, c) \in \mathcal{B}(\mathbf{C})$, let

$$E_b: \ y^2 = x^3 + Ax + B, \qquad C_b: \ w^3 = y - c.$$

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Here C_b denotes the smooth projective cyclic cover defined by the section $y - c \in H^0(E_b, \mathcal{O}_{E_b}(3O))$; the displayed equation is its affine chart. The reduced branch divisor makes this cover connected. Let

$$\pi_b : C_b \longrightarrow E_b, \qquad (x, y, w) \longmapsto (x, y)$$

be the cyclic triple cover. The condition $\Delta_{\mathrm{br}} \neq 0$ means that the line $y = c$ meets E_b in three distinct points; these are precisely the branch points of π_b. Write $O \in E_b$ for the point at infinity and set

$$\kappa_b = \pi_b^* \mathcal{O}_{E_b}(O), \qquad P_b = \ker(\mathrm{Nm}_{\pi_b})^0.$$

The centred theta divisor on $J(C_b)$ restricts to the divisor

$$X_b = \{M \in P_b : h^0(C_b, \kappa_b \otimes M) > 0\}.$$

Write $\mathcal{P} \to \mathcal{B}$ and $\mathcal{X} \subset \mathcal{P}$ for the resulting relative Prym and relative theta section.

Theorem 1.1 (Main theorem)**.** *For every $b \in \mathcal{B}(\mathbf{C})$ the following hold.*

(i) *P_b is an abelian threefold, and the polarization induced by the centred theta divisor has type $(1, 1, 3)$.*

(ii) *$X_b \subset P_b$ is an integral normal symmetric ample Cartier divisor with a unique singular point at the origin; this singularity is simple elliptic of type $\widetilde{E}_7$.*

(iii) *$\chi(\mathrm{IC}_{X_b}) = 7$, and the associated Tannaka pair satisfies*

$$\big(G(\mathrm{IC}_{X_b}), \omega(\mathrm{IC}_{X_b})\big) \simeq (G_2, V_7),$$

where V_7 is the standard seven-dimensional representation of G_2. In particular, the full convolution Tannaka group is connected and isomorphic to G_2.

Moreover, $\mathcal{X} \subset \mathcal{P}$ is a symmetric relatively ample effective Cartier divisor, flat over $\mathcal{B}$, with geometrically integral fibres.

Proof outline. Fix $b \in \mathcal{B}(\mathbf{C})$ and write $(P, X) = (P_b, X_b)$. The proof has a geometric part and a Tannakian part, with the pair-incidence calculation linking the two. The norm fibre in $C^{(3)}$ is birational to X and has one $\widetilde{E}_6$ singularity. Blowing up that point and contracting the strict transform of the trigonal pencil gives the minimal resolution $\rho : S \to X$; its exceptional locus is an elliptic curve of self-intersection -2. Adjunction, the local geometric genus, and Noether's formula give

$$K_S^2 = 16, \qquad \chi(\mathcal{O}_S) = 2, \qquad e(S) = 8,$$

while the theta group yields $p_g(S) = 5$ and $q(S) = 4$. The semismall decomposition then gives $\chi(\mathrm{IC}_X) = 7$, and the unique singular point forces $\mathrm{Stab}_P(X) = \{0\}$. A generic unitary twist of the same resolution has Hodge multiplicities $(2, 3, 2)$.

The geometric core of the argument is the generic pair-incidence fibre. For a geometric generic point x of $\mathcal{M} = (\mathcal{X}/\mathcal{B})_{\mathrm{sm}}$, let D_x be its unique effective divisor of degree three. The cyclic norm of a section of $H^0(C, 3\kappa - D_x)$ is the fixed norm line times a conic. Its determinant has a structural linear factor and a geometrically integral residual octic. The factor-ordering cover on that octic is split directly by the cubic norm: after restricting to the line $Y = (-v/z)^3$, the residual conic is the product of the two nontrivial cubic conjugates of the section. This gives two rationally labelled factors over $\mathbf{C}$. The scheme-theoretic Abel inverse converts them into the two nonexceptional components of

$$C_x = \{y \in X : -x - y \in X\},$$

and a separate boundary argument shows that no further nonexceptional component occurs.

After spreading these two components over a nonempty open $\mathcal{M}^\circ \subset \mathcal{M}$, top cohomology of the pair fibres and the decomposition theorem produce an unshifted summand $\mathrm{IC}_X \subset \mathrm{Alt}^2(\mathrm{IC}_X)$. If $G = G(\mathrm{IC}_X)$ and $V = \omega(\mathrm{IC}_X)$, the final input is therefore

$$G \subset \mathrm{O}(V), \qquad \dim V = 7, \qquad V|_{G^0} \text{ irreducible},$$
$$\mu : \mathbf{G}_m \longrightarrow [G^0, G^0] \quad \text{with weights } -2, 0, 2 \text{ of multiplicities } 2, 3, 2,$$
$$\mathrm{Hom}_G(V, \Lambda^2 V) \neq 0.$$

The exterior-square morphism rules out SO_7, the Hodge cocharacter rules out the seven-dimensional PGL_2 module, and the same morphism excludes the only possible disconnected extension of G_2. Thus $G = G_2$.

This argument first applies on a nonempty open $\mathcal{B}^\circ$. Blowing up the singular section of the relative norm fibre gives a smooth proper family resolving every X_b. Its semismall decomposition shows that nearby cycles preserve IC_{X_b}. Specialization embeds each remaining Tannaka group into the generic G_2; connected irreducibility and the same Hodge multiplicities force equality (Proposition 8.1 and Theorem 8.2).

Unless a rational model or a finite-field reduction is explicitly mentioned, the geometric construction is over $\mathbf{C}$. Intersection complexes, cohomology, and Hodge modules retain their canonical $\mathbf{Q}$-structures. The Tannakian category is obtained after extension of the underlying perverse sheaves from $\mathbf{Q}$ to $\mathbf{C}$. Thus all Tannaka groups and tautological modules below are over $\mathbf{C}$. The convention $\mathbf{P}(\mathcal{E}) = \mathrm{Proj}(\mathrm{Sym}\, \mathcal{E}^\vee)$ is used, so $\mathbf{P}(\mathcal{E})$ parametrizes lines in the fibres of $\mathcal{E}$. For an integral scheme Y, $k(Y)$ denotes its function field and $k(y)$ the residue field of a point y.

2. Tannakian conventions

Let A be a complex abelian variety and let $m : A \times A \to A$ be addition. For constructible complexes with coefficients in $\mathbf{Q}$ or $\mathbf{C}$, set

$$K * L = Rm_*(K \boxtimes L).$$

Set

$$\mathsf{S}(A) = \{Q \in \mathrm{Perv}(A, \mathbf{C}) : \chi(A, Q) = 0\},$$
$$\mathsf{T}(A) = \{M \in D^b_c(A, \mathbf{C}) : {}^pH^i(M) \in \mathsf{S}(A) \text{ for every } i\}.$$

The first is a Serre subcategory and the second is the corresponding triangulated tensor ideal. The quotient

$$\overline{D}(A) = D^b_c(A, \mathbf{C})/\mathsf{T}(A)$$

inherits the perverse t-structure; its heart is

$$\overline{\mathrm{Perv}}(A) = \mathrm{Perv}(A, \mathbf{C})/\mathsf{S}(A).$$

Convolution is t-exact on $\overline{D}(A)$, and the displayed heart is a rigid symmetric monoidal neutral Tannakian category [9, Proposition 3.1]. For the pure polarizable objects used below, the tensor subcategory generated by one object is semisimple by the decomposition theorem; its Tannaka group is therefore reductive. Let δ_0 denote the skyscraper perverse sheaf at the origin; its image is the tensor unit for convolution.

For an integral subvariety $Z \subset A$ of dimension d, let

$$\mathrm{IC}_Z = j_{!*}\mathbf{Q}_{Z_{\mathrm{reg}}}[d], \qquad j : Z_{\mathrm{reg}} \hookrightarrow A,$$

be its perverse intersection complex, extended by zero to A. If $\chi(\mathrm{IC}_Z) > 0$, let $G(\mathrm{IC}_Z)$ denote the Tannaka group of the category generated by $\mathrm{IC}_Z \otimes_{\mathbf{Q}} \mathbf{C}$ in the displayed quotient, and by $\omega(\mathrm{IC}_Z)$ its tautological module. Then

$$\dim \omega(\mathrm{IC}_Z) = \chi(\mathrm{IC}_Z).$$

The action on the tautological module is faithful by construction. Scalar extension is suppressed inside Tannakian expressions. Let IC_Z^H denote the polarizable pure Hodge module whose underlying perverse $\mathbf{Q}$-sheaf is IC_Z.

The commutativity constraint contains the usual cohomological sign. Since the support in this paper has even dimension two, the alternating projector on $\mathrm{IC}_X * \mathrm{IC}_X$ is carried by the fibre functor to the ordinary exterior square of $\omega(\mathrm{IC}_X)$. Let $\mathrm{Alt}^2(\mathrm{IC}_X)$ denote its image in the Tannakian quotient.

3. The shifted cyclic theta section

Until the beginning of Section 4, fix $b = (A, B, c) \in \mathcal{B}(\mathbf{C})$ and suppress it from the notation. Thus

$$E : y^2 = f_3(x), \qquad C : w^3 = y - c,$$

where $f_3(x) = x^3 + Ax + B$. Set

$$Y = y - c,$$

so that $w^3 = Y$. The map $\pi : C \to E$ is a cyclic cover of degree three. Its branch divisor is the reduced intersection of E with the line $y = c$.

3.1. **The cover, the Prym, and the polarization.** Let $b_1, b_2, b_3 \in E$ be the branch points and let $r_i \in C$ be the ramification point above b_i. Since

$$\mathrm{div}_E(y - c) = b_1 + b_2 + b_3 - 3O,$$

the branch divisor belongs to $|3O|$ and, in the group law of E,

$$b_1 + b_2 + b_3 = O. \tag{3.1}$$

Put $\eta = \mathcal{O}_E(O)$. The cyclic covering algebra is

$$\pi_* \mathcal{O}_C = \mathcal{O}_E \oplus \eta^{-1} \oplus \eta^{-2}.$$

Riemann–Hurwitz gives $g(C) = 4$, and the canonical bundle formula gives

$$K_C = \pi^*(K_E \otimes \eta^2) = \pi^* \eta^2 = \kappa^2, \qquad \kappa = \pi^* \eta.$$

By projection formula,

$$h^0(C, \kappa) = h^0(E, \eta) + h^0(E, \mathcal{O}_E) + h^0(E, \eta^{-1}) = 2.$$

The pullback $\pi^* : J(E) \to J(C)$ is injective. Indeed, if $\alpha \in \mathrm{Pic}^0(E)$ and $\pi^* \alpha \simeq \mathcal{O}_C$, then

$$1 = h^0(C, \pi^* \alpha) = h^0(E, \alpha) + h^0(E, \alpha \eta^{-1}) + h^0(E, \alpha \eta^{-2}),$$

and the last two terms vanish; hence $\alpha = \mathcal{O}_E$. By duality, $\ker(\mathrm{Nm}_\pi)$ is connected. Thus set

$$P = \ker(\mathrm{Nm}_\pi).$$

It has dimension three. If $\tau(w) = \zeta_3 w$, then $\pi^* \operatorname{Nm}_\pi = 1 + \tau + \tau^2$ on $J(C)$. Consequently $\operatorname{Im}(1-\tau) \subset P$. The τ-invariant part of $H^0(C, \Omega^1_C)$ is $\pi^* H^0(E, \Omega^1_E)$ and has dimension one, so $\operatorname{Im}(1-\tau)$ has dimension three. Hence

$$P = \operatorname{Im}(1-\tau) = \ker(\operatorname{Nm}_\pi). \tag{3.2}$$

The same argument works in the relative family. Indeed $\operatorname{Nm}_\pi \circ \pi^* = [3]$ on the relative elliptic Jacobian, so the norm map is surjective; its kernel is smooth, and the fibrewise injectivity of π^* proved above makes every kernel fibre connected. Thus $\mathcal{P} = \ker(\operatorname{Nm}_\pi) \to \mathcal{B}$ is an abelian scheme of relative dimension three. Use the theta characteristic κ to centre the theta divisor on $J(C) = \operatorname{Pic}^0(C)$:

$$\Theta_\kappa = \{M \in J(C) : h^0(C, \kappa \otimes M) > 0\}.$$

Let $L = \mathcal{O}_{J(C)}(\Theta_\kappa)|_P$. The type of the restricted polarization can be read off directly. Under the principal polarizations of the two Jacobians, π^* and Nm_π are dual homomorphisms; hence the orthogonal complement of P in $J(C)$ is $\pi^* J(E)$ [2, Section 12.3]. Therefore

$$\ker(\varphi_L) = P \cap \pi^* J(E).$$

Since π^* is injective and $\operatorname{Nm}_\pi \circ \pi^* = [3]$ on $J(E)$,

$$P \cap \pi^* J(E) = \pi^* J(E)[3].$$

This group has order 9. If the polarization type is (d_1, d_2, d_3) with $d_1 \mid d_2 \mid d_3$, then $|\ker(\varphi_L)| = (d_1 d_2 d_3)^2$; hence $d_1 d_2 d_3 = 3$ and necessarily

$$\operatorname{type}(L) = (1,1,3), \qquad L^3 = 3!\, d_1 d_2 d_3 = 18. \tag{3.3}$$

Scheme-theoretically,

$$X = \Theta_\kappa \cap P = \{M \in P : h^0(C, \kappa \otimes M) > 0\}.$$

The theta divisor is preserved as a closed subscheme by the Serre-duality involution $L' \mapsto K_C \otimes (L')^{-1}$ on $\operatorname{Pic}^3(C)$. Since $\kappa^2 = K_C$, this involution is $M \mapsto M^{-1}$ in centred coordinates. Relative Serre duality preserves the determinantal ideal of theta, so after restriction to P one obtains

$$X = -X \tag{3.4}$$

as closed subschemes. Equation (3.8) will show that $X = a - X$ implies $a = 0$.

3.2. **The norm fibre and the resolution.** Let

$$n : C^{(3)} \longrightarrow \operatorname{Pic}^3(E), \qquad D \longmapsto \mathcal{O}_E(\pi_* D),$$

and put

$$N = n^{-1}(\eta^3), \qquad a : N \longrightarrow X, \qquad D \longmapsto \mathcal{O}_C(D) \otimes \kappa^{-1}.$$

For the centred Abel map $D \mapsto \mathcal{O}_C(D) \otimes \kappa^{-1}$ one has scheme-theoretically

$$N = C^{(3)} \times_{J(C)} P.$$

Thus a is its base change to P. The map a is surjective: if $M \in X$, a nonzero section of $\kappa \otimes M$ has an effective divisor D of degree three, and

$$\operatorname{Nm}_\pi \mathcal{O}_C(D) = \operatorname{Nm}_\pi(\kappa \otimes M) = \eta^3.$$

The finite morphism $C^{(3)} \to E^{(3)}$ induced by π restricts to a finite surjection

$$N \longrightarrow |\eta^3| \simeq \mathbf{P}^2.$$

Hence $\dim N = 2$, so the restriction of the centred theta section to P is nonzero. Therefore $X \in |L|$ is an effective Cartier divisor, and it is ample.

Let $\Gamma = |\kappa| \simeq \mathbf{P}^1$ and put

$$D_0 = r_1 + r_2 + r_3 \in \Gamma.$$

Lemma 3.1 (The norm fibre)**.** *The surface N has the unique singular point D_0, and the germ (N, D_0) is a simple elliptic singularity of type $\widetilde{E}_6$. If $\beta : \widehat{N} \to N$ is the blowup of D_0, then $\widehat{N}$ is smooth, its exceptional curve E_0 is elliptic with $E_0^2 = -3$, and the strict transform $\widetilde{\Gamma}$ is a (-1)-curve meeting E_0 transversely in one point. Contracting $\widetilde{\Gamma}$ gives a smooth surface S and a proper morphism*

$$\rho : S \longrightarrow X.$$

The image E_{exc} of E_0 satisfies $E_{\mathrm{exc}}^2 = -2$ and maps to the origin of X.

Proof. Because the three branch points are distinct, a neighbourhood of D_0 in $C^{(3)}$ is the product of neighbourhoods of the r_i. Choose additive analytic coordinates t_i on E at b_i so that the Abel map $E^{(3)} \to \mathrm{Pic}^3(E)$ is, after translation, given by $t_1 + t_2 + t_3$. After rescaling local parameters z_i at r_i, the cover has $t_i = z_i^3$. Hence the norm fibre has the analytic equation

$$z_1^3 + z_2^3 + z_3^3 = 0.$$

Thus (N, D_0) is the affine cone over a smooth plane cubic, hence a simple elliptic singularity of type $\widetilde{E}_6$.

It remains to locate the other critical points of the norm map. Separate the possible multiplicity patterns on the symmetric cube. A divisor containing an unramified point has a tangent direction on which $d\pi$ is nonzero, so it cannot be critical. Near a ramification point the map is $z \mapsto z^3$. If two points collide, then in the elementary symmetric coordinates $e_1 = z_1 + z_2$ and $e_2 = z_1 z_2$ one has

$$z_1^3 + z_2^3 = e_1^3 - 3e_1e_2,$$

which has no linear term at the double ramification cycle. If three points collide, however,

$$z_1^3 + z_2^3 + z_3^3 = e_1^3 - 3e_1e_2 + 3e_3,$$

and the $3e_3$ term makes the differential nonzero. Thus, apart from the cycle $D_0 = r_1 + r_2 + r_3$, the only possible critical cycles are $2r_i + r_j$ with $i \neq j$. Membership in the norm fibre would then force

$$2b_i + b_j \sim 3O.$$

If k is the third index, (3.1) gives $b_i + b_j + b_k \sim 3O$, hence $b_i = b_k$, contradicting the fact that the branch points are distinct. Therefore D_0 is the only singular point of N.

Blowing up the vertex of the cubic cone gives a smooth surface $\widehat{N}$ with exceptional elliptic curve E_0 and $E_0^2 = -3$. The local behaviour of the pencil $\Gamma = |\kappa|$ at the vertex is as follows. Projection formula gives

$$\pi_*\kappa \simeq \eta \oplus \mathcal{O}_E \oplus \eta^{-1}.$$

The two sections of κ may therefore be chosen as follows: the section coming from the $\mathcal{O}_E$-summand is represented by w and has zero divisor D_0, while a generator of $H^0(E, \eta)$ pulls back to a section that is nonzero at each r_i, since none of the branch

points b_i is O. Let t be the affine parameter of the pencil centred at the member D_0. In a local trivialization of κ at r_i, the moving section has equation

$$w + tu_i = 0, \qquad u_i(r_i) \neq 0.$$

Since w is a local parameter at the totally ramified point r_i, after rescaling z_i this gives

$$z_i(t) = -u_i(r_i)t + O(t^2).$$

Thus $\Gamma \to N$ has a nonzero tangent vector at D_0 and hence multiplicity one there. Because $\Gamma \subset N$, its tangent direction lies on the exceptional cubic

$$\lambda_1^3 + \lambda_2^3 + \lambda_3^3 = 0, \qquad \lambda_i = -u_i(r_i),$$

and the displayed expansion shows that the strict transform of Γ meets E_0 in one reduced point, transversely.

Let $a_C : C^{(3)} \to J(C)$ be the Abel–Jacobi map and let Θ_C be the principal theta divisor. If $X_{p_0} \subset C^{(d)}$ denotes the divisor of cycles containing a fixed point p_0, Macdonald's canonical-class formula is [16]

$$K_{C^{(d)}} \equiv a_d^* \Theta_C + (g - d - 1)X_{p_0}.$$

For $g(C) = 4$ and $d = 3 = g - 1$ the second term vanishes, hence

$$K_{C^{(3)}} \equiv a_C^* \Theta_C.$$

Since N is a fibre of $n : C^{(3)} \to \mathrm{Pic}^3(E)$, its normal line bundle is trivial. Adjunction therefore gives

$$K_N \equiv a^* L.$$

The map a contracts Γ, so $K_N \cdot \Gamma = 0$. The cubic cone has discrepancy -1. Here N is a Cartier divisor in the smooth variety $C^{(3)}$, hence is Gorenstein. The numerical equivalence above is used only for intersection numbers, while the local cubic-cone equation gives the divisor identity

$$K_{\widehat{N}} = \beta^* K_N - E_0.$$

Since $E_0 \cdot \widetilde{\Gamma} = 1$, one has $K_{\widehat{N}} \cdot \widetilde{\Gamma} = -1$. Adjunction on $\widetilde{\Gamma} \simeq \mathbf{P}^1$ then gives $\widetilde{\Gamma}^2 = -1$. Contracting it produces a smooth surface S and

$$E_{\mathrm{exc}}^2 = E_0^2 + (E_0 \cdot \widetilde{\Gamma})^2 = -2.$$

The composite $\widehat{N} \to N \to X$ is constant on $\widetilde{\Gamma}$, hence factors through $\rho : S \to X$. The curve E_{exc} maps to the origin. □

Proposition 3.2 (The theta surface and its resolution)**.** *The theta section $X \subset P$ is an integral normal symmetric ample Cartier divisor with* $\mathrm{Sing}(X) = \{0\}$*. The morphism $\rho : S \to X$ of Lemma 3.1 is its minimal resolution and is an isomorphism over $X \setminus \{0\}$. Its exceptional locus is the smooth elliptic curve E_{exc} with $E_{\mathrm{exc}}^2 = -2$; in particular, $(X, 0)$ is a simple elliptic singularity of type $\widetilde{E}_7$.*

Proof. A basis of $H^0(C, K_C)$ is represented on the affine chart by

$$\frac{dx}{y}, \qquad \frac{dx}{wy}, \qquad \frac{dx}{w^2 y}, \qquad \frac{x\,dx}{w^2 y}.$$

Indeed these forms are regular both at the ramification points and at the three points above O. After multiplication by the common factor $w^2 y/dx$, the canonical map is

$$(x, w) \longmapsto [w^2 : w : 1 : x].$$

The map is birational, so C is not hyperelliptic. Among the ten quadratic monomials in the four canonical coordinates there is the evident relation $X_0X_2 - X_1^2 = 0$, and it is the only one. After restriction to C, the other nine monomials may be represented as

$$\underbrace{1, x, Y, x^2}_{\tau\text{-eigenvalue } 1}\ ; \qquad \underbrace{w, xw, wY}_{\tau\text{-eigenvalue } \zeta_3}\ ; \qquad \underbrace{w^2, xw^2}_{\tau\text{-eigenvalue } \zeta_3^2}\ .$$

At any of the three points above O their pole orders within these three eigenspaces are, respectively,

$$(0, 2, 3, 4), \qquad (1, 3, 4), \qquad (2, 4).$$

They are therefore linearly independent within each eigenspace, and terms from distinct eigenspaces cannot cancel. Thus the canonical ideal has a single quadric,

$$X_0X_2 - X_1^2 = 0,$$

which has rank three and one ruling. Every degree-three line bundle A with $h^0(C, A) \geq 2$ is base-point-free: otherwise, for a base point p, the degree-two line bundle $A(-p)$ would have at least two sections and would define a g^1_2, contrary to nonhyperellipticity. Geometric Riemann–Roch therefore identifies every g^1_3 with a ruling of the canonical quadric. Its unique ruling is the one cut out by $|\kappa|$, so κ is the unique g^1_3 on C. Riemann–Kempf [11] shows that the Abel map is an isomorphism over the locus $h^0 = 1$. The corresponding points of N are smooth by the critical-point calculation above, and $|\kappa|$ is the only positive-dimensional Abel fibre. It follows that $\mathrm{Sing}(X) \subset \{0\}$.

The Riemann singularity theorem [11] gives $\mathrm{mult}_0(\Theta_\kappa) = h^0(C, \kappa) = 2$. The restricted theta equation is nonzero, since $\dim X = 2 < \dim P$, and lies in $\mathfrak{m}^2_{P,0}$. Thus X is singular at the origin, and

$$\mathrm{Sing}(X) = \{0\}. \tag{3.5}$$

The Cartier divisor X is generically reduced and Cohen–Macaulay, hence reduced. The divisor sequence, together with Kodaira vanishing and Serre duality for L^{-1}, gives $H^0(X, \mathcal{O}_X) = \mathbf{C}$; thus X is connected. Suppose that X were reducible. Two irreducible components joined by an edge in the incidence graph must then meet. Both have dimension two in the smooth threefold P, so every nonempty intersection of two distinct components has dimension at least one. At such a point the local ring of X has at least two minimal primes and is therefore not regular. Hence the intersection is contained in $\mathrm{Sing}(X) = \{0\}$, contradicting its positive dimension. Thus X is integral. Being Cohen–Macaulay, it satisfies S_2, while (3.5) gives regularity in codimension one. Serre's criterion makes X normal.

The uniqueness of the g^1_3 shows that every $M \neq 0$ in X satisfies $h^0(C, \kappa \otimes M) = 1$, whereas the Abel fibre over 0 is precisely $\Gamma = |\kappa|$. Base change of the Abel map therefore gives an isomorphism

$$N \setminus \Gamma \xrightarrow{\sim} X \setminus \{0\}.$$

The surface N is reduced: it is smooth away from D_0, and its germ at D_0 is the integral cubic cone. It is also a Cartier divisor in the smooth variety $C^{(3)}$, hence is pure of dimension two; no irreducible component can be contained in the curve Γ. The displayed integral dense open then shows that N is integral. Consequently ρ is a proper birational resolution whose only exceptional curve is E_{exc}. Since this curve

is elliptic with self-intersection -2, the resolution is minimal and the simple-elliptic classification identifies the germ with type $\widetilde{E}_7$ [14, 17]. □

Lemma 3.3 (Surface invariants)**.** *The minimal resolution S satisfies*

$$K_S^2 = 16, \qquad \chi(\mathcal{O}_S) = 2, \qquad e(S) = 8.$$

Proof. From $K_N \equiv a^*L$ and $X \in |L|$ one obtains

$$K_N^2 = (a^*L)^2 = L^2 \cdot [X] = L^3 = 18.$$

For the blowup of the $\widetilde{E}_6$ vertex, $K_{\widehat{N}} = \beta^* K_N - E_0$, so

$$K_{\widehat{N}}^2 = 18 + E_0^2 = 15.$$

The contraction $\widehat{N} \to S$ is the inverse of a blowup at a smooth point, and therefore $K_S^2 = 16$.

The Cartier sequence

$$0 \longrightarrow L^{-1} \longrightarrow \mathcal{O}_P \longrightarrow \mathcal{O}_X \longrightarrow 0$$

gives $\chi(\mathcal{O}_X) = 3$: on the abelian threefold P, $\chi(\mathcal{O}_P) = 0$, $\chi(L) = 3$, and $\chi(L^{-1}) = -\chi(L) = -3$. A simple elliptic singularity has local geometric genus one; equivalently, $R^1\rho_*\mathcal{O}_S$ is a skyscraper sheaf of length one [14]. Since X is normal, $\rho_*\mathcal{O}_S = \mathcal{O}_X$, and Leray gives

$$\chi(\mathcal{O}_S) = \chi(\mathcal{O}_X) - 1 = 2.$$

Noether's formula now yields

$$e(S) = 12\chi(\mathcal{O}_S) - K_S^2 = 24 - 16 = 8.$$

□

Corollary 3.4 (The intersection complex)**.** *There is a decomposition*

$$R\rho_*\mathbf{Q}_S[2] \simeq \mathrm{IC}_X \oplus \delta_0.$$

Moreover,

$$\chi(\mathrm{IC}_X) = 7, \qquad \mathrm{Stab}_P(X) = \{0\}.$$

Proof. The map $\rho : S \to X$ is semismall, with the single relevant exceptional fibre E_{exc}. Its top intersection form is the nondegenerate matrix (-2). The semismall decomposition theorem [4, Theorem 3.4.1] gives

$$R\rho_*\mathbf{Q}_S[2] \simeq \mathrm{IC}_X \oplus \delta_0. \tag{3.6}$$

Taking Euler characteristics and using Lemma 3.3 gives

$$\chi(\mathrm{IC}_X) = e(S) - 1 = 7. \tag{3.7}$$

A translation preserving X must preserve its unique singular point, hence

$$\mathrm{Stab}_P(X) = \{0\}. \tag{3.8}$$

In particular, $X = a - X$ holds only for $a = 0$. □

Let $\Lambda_X \subset T^*P$ be the closure of the conormal bundle of X_{reg}, and let $\Lambda_0 = T_0^*P$ be the conormal fibre at the origin. Let

$$\gamma_X : \mathbf{P}(\Lambda_X) \longrightarrow \mathbf{P}(T_0^*P)$$

be the map induced by cotangent projection.

Lemma 3.5 (Conormal degree)**.** *The characteristic cycle and conormal Gauss degree are*

$$\mathrm{CC}(\mathrm{IC}_X) = \Lambda_X + \Lambda_0, \qquad \deg(\gamma_X) = 6.$$

In particular, the conormal Gauss map is dominant.

Proof. The local Euler obstruction is invariant under analytic isomorphism, so it may be computed in the standard hypersurface embedding of a $\widetilde{E}_7$ simple-elliptic germ,

$$z^2 + f_4(x, y) = 0,$$

where f_4 is a square-free binary quartic [14, 17]. In these coordinates take a general linear form $\ell = z - ax - by$. On $\ell = 0$ the quadratic term $(ax + by)^2$ has rank one, and square-freeness makes the restriction of f_4 to its null line nonzero for general $(a : b)$. The analytic splitting lemma therefore reduces the plane-curve germ to $s^2 + t^4 = 0$, namely type A_3, whose Milnor number is 3. The Lefschetz formula for the local Euler obstruction [3, Theorem 3.1] identifies $\mathrm{Eu}_X(0)$ with the Euler characteristic of this generic linear Milnor fibre. Hence

$$\mathrm{Eu}_X(0) = 1 - \mu(A_3) = 1 - 3 = -2.$$

On the other hand, the stalk of $R\rho_*\mathbf{Q}_S[2]$ at the origin has Euler characteristic $e(E_{\mathrm{exc}}) = 0$, whereas the summand δ_0 has stalk Euler characteristic 1. Thus (3.6) gives $\chi((\mathrm{IC}_X)_0) = -1$. The Dubson–Kashiwara local index formula [10] therefore gives

$$\mathrm{CC}(\mathrm{IC}_X) = \Lambda_X + m\Lambda_0, \qquad -1 = -2 + m,$$

hence $m = 1$. The Gauss map of $\Lambda_0 = T_0^*P$ is the identity on $\mathbf{P}(T_0^*P)$ and has degree one. The Franecki–Kapranov index theorem [5, Theorem 1.3], together with (3.7), then gives

$$7 = \deg(\gamma_X) + 1.$$

Thus

$$\deg(\gamma_X) = 6. \tag{3.9}$$

□

3.3. **The irregularity and the generic-twist Hodge vector.** The exceptional elliptic curve contributes to $H^1(S)$. Its contribution is determined before introducing the generic twist.

Proposition 3.6 (Irregularity of the resolution)**.** *The minimal resolution S satisfies*

$$p_g(S) = 5, \qquad q(S) = 4.$$

Consequently

$$b_1(S) = 8, \qquad b_2(S) = 22, \qquad h^{1,1}(S) = 12.$$

Proof. Recall from (3.3) that

$$K(L) := \ker(\varphi_L) = \pi^*E[3]$$

has order nine. Every point of $K(L)$ lies on X. For $\alpha \in E[3]$, the projection formula and $\pi_*\mathcal{O}_C = \mathcal{O}_E \oplus \eta^{-1} \oplus \eta^{-2}$ give

$$h^0(C, \kappa \otimes \pi^*\alpha) = h^0(E, \eta \otimes \alpha) + h^0(E, \alpha) + h^0(E, \eta^{-1} \otimes \alpha).$$

The first term is one because $\deg \eta = 1$; the other two vanish when $\alpha \neq \mathcal{O}_E$. For $\alpha = \mathcal{O}_E$ the same expression is two. Thus

$$K(L) \subset X. \tag{3.10}$$

Let $s \in H^0(P, L)$ be a defining section of X. The theta group of L acts on the three-dimensional space $H^0(P, L)$ through its irreducible Schrödinger representation; see, for example, [2, Chapter 6]. For $a \in K(L)$ choose a lift of translation by a to the theta group. The corresponding translate of s vanishes at the origin, because its value there is a nonzero scalar multiple of $s(\pm a)$, depending on the action convention; both values vanish by (3.4) and (3.10). The orbit of the nonzero vector s spans $H^0(P, L)$ by irreducibility. Hence

$$t(0) = 0 \qquad \text{for every } t \in H^0(P, L). \tag{3.11}$$

In other words, the origin is a base point of the complete linear system $|L|$.

Adjunction gives $\omega_X \simeq L|_X$, and the divisor sequence is

$$0 \longrightarrow \mathcal{O}_P \xrightarrow{s} L \longrightarrow \omega_X \longrightarrow 0.$$

Since L is ample on the abelian threefold, $H^i(P, L) = 0$ for $i > 0$. Therefore

$$0 \longrightarrow \mathbf{C}s \longrightarrow H^0(P, L) \longrightarrow H^0(X, \omega_X) \longrightarrow H^1(P, \mathcal{O}_P) \longrightarrow 0, \tag{3.12}$$

so $h^0(X, \omega_X) = 2 + 3 = 5$.

All five canonical sections vanish at the singular point. The two-dimensional subspace coming from $H^0(P, L)/\mathbf{C}s$ does so by (3.11). For the remaining three directions, let v be a translation-invariant vector field on P. Choose local frames of L with transition functions g_{ij} and write the defining section as local functions s_i, so $s_i = g_{ij} s_j$. Differentiating gives

$$v(s_i) - g_{ij} v(s_j) = v(g_{ij}) s_j.$$

The right-hand side vanishes on X, so the restrictions of the $v(s_i)$ glue to a well-defined section $\partial_v s \in H^0(X, L|_X)$. Under the boundary map in (3.12), this section maps, up to the usual sign convention, to

$$\iota_v c_1(L) \in H^1(P, \mathcal{O}_P),$$

because the associated Čech cocycle is $v(g_{ij})/g_{ij}$. This is exactly the differential at the origin of the polarization morphism $\varphi_L : P \to \mathrm{Pic}^0(P)$. Since φ_L is an isogeny,

$$H^0(P, T_P) \longrightarrow H^1(P, \mathcal{O}_P), \qquad v \longmapsto \iota_v c_1(L),$$

is an isomorphism. The three sections $\partial_v s$ therefore map onto the last term of (3.12). They also vanish at the origin because 0 is a singular point of X, so $ds(0) = 0$. Thus every section of ω_X vanishes at 0.

Since K_X is Cartier, write $K_S = \rho^* K_X + a E_{\mathrm{exc}}$. Adjunction on the exceptional elliptic curve gives $(K_S + E_{\mathrm{exc}}) \cdot E_{\mathrm{exc}} = 0$, while $\rho^* K_X \cdot E_{\mathrm{exc}} = 0$ and $E_{\mathrm{exc}}^2 = -2$. Thus $a = -1$ and

$$K_S = \rho^* K_X - E_{\mathrm{exc}}. \tag{3.13}$$

The ideal appearing after resolution is also explicit. Since X is normal and ρ is a resolution, $\rho_* \mathcal{O}_S = \mathcal{O}_X$. Push forward

$$0 \longrightarrow \mathcal{O}_S(-E_{\mathrm{exc}}) \longrightarrow \mathcal{O}_S \longrightarrow \mathcal{O}_{E_{\mathrm{exc}}} \longrightarrow 0.$$

The exceptional curve is connected and projective, so its global regular functions are constant. Since $\rho(E_{\mathrm{exc}}) = \{0\}$, the induced map

$$\mathcal{O}_X \longrightarrow \rho_*\mathcal{O}_{E_{\mathrm{exc}}}$$

is evaluation at the reduced point 0 and has kernel $\mathfrak{m}_0$. Left exactness of ρ_* gives

$$\rho_*\mathcal{O}_S(-E_{\mathrm{exc}}) = \mathfrak{m}_0.$$

By the projection formula,

$$H^0(S, K_S) = H^0(X, \omega_X \otimes \mathfrak{m}_0).$$

The preceding paragraph shows that the right-hand side is all of $H^0(X, \omega_X)$, and therefore $p_g(S) = 5$. Finally Lemma 3.3 gives $\chi(\mathcal{O}_S) = 2$, so

$$q(S) = 1 + p_g(S) - \chi(\mathcal{O}_S) = 4.$$

The stated Betti and Hodge numbers now follow from $e(S) = 8$. □

Let $j : S \xrightarrow{\rho} X \hookrightarrow P$. A unitary rank-one character of $\pi_1(P)$ determines a unitary local system $\mathcal{L}_\alpha$ and a holomorphic line bundle $\alpha \in \mathrm{Pic}^0(P)$. The same symbol α will also denote the latter.

From this point, IC_X^H also denotes its scalar extension to a polarizable complex Hodge module. A unitary rank-one local system underlies a unique polarizable complex variation of Hodge structure of weight zero and type $(0,0)$. Thus twisting by $\mathcal{L}_\alpha$ is defined for polarizable complex Hodge modules, and its cohomology has a functorial complex Hodge decomposition [12, Sections 2.1 and 2.5]. For $\alpha \neq \mathcal{O}_P$, the standard vanishing for a nontrivial line bundle in $\mathrm{Pic}^0(P)$ and the divisor sequence give

$$H^0(X, (L \otimes \alpha)|_X) \simeq H^0(P, L \otimes \alpha), \qquad h^0(P, L \otimes \alpha) = 3.$$

Combining this with (3.13) and $\rho_*\mathcal{O}_S(-E_{\mathrm{exc}}) = \mathfrak{m}_0$ yields

$$H^0(S, K_S \otimes j^*\alpha) \simeq \ker\Big(H^0(P, L \otimes \alpha) \xrightarrow{\mathrm{ev}_0} (L \otimes \alpha)_0\Big). \tag{3.14}$$

The locus on which the evaluation map in (3.14) is nonzero is a nonempty Zariski-open subset of $\mathrm{Pic}^0(P)$. For nonemptiness, use the surjectivity of φ_L: every $L \otimes \alpha$ is a translate of L, and a translate of the defining section by a point outside X does not vanish at the origin. Intersecting this open with its inverse under $\alpha \mapsto \alpha^{-1}$ gives a dense open $W \subset \mathrm{Pic}^0(P)$ on which

$$h^0(S, K_S \otimes j^*\alpha) = h^0(S, K_S \otimes j^*\alpha^{-1}) = 2. \tag{3.15}$$

Let U_{GV} be the nonempty Zariski-open subset of the Betti character torus supplied by perverse generic vanishing, on which

$$H^k(P, Rj_*\mathbf{Q}_S[2] \otimes \mathcal{L}_\alpha) = 0 \qquad (k \neq 0).$$

The compact unitary character torus is Zariski dense in the Betti character torus, so its intersection with U_{GV} is dense and Euclidean open. The unitary-character correspondence

$$\mathrm{Char}_{\mathrm{u}}(P) \xrightarrow{\sim} \mathrm{Pic}^0(P)$$

is a real-analytic isomorphism; the inverse image of the dense Zariski-open set W is therefore also dense and Euclidean open. Their intersection is a nonempty Euclidean open subset Ω of the unitary character torus.

Proposition 3.7 (Generic-twist Hodge vector)**.** *For every $\alpha \in \Omega$, the complex Hodge decomposition on $H^0(P, \mathrm{IC}_X^H \otimes \mathcal{L}_\alpha)$ has Hodge numbers*

$$(h^{2,0}, h^{1,1}, h^{0,2}) = (2, 3, 2).$$

Equivalently, the associated antidiagonal Hodge cocharacter has weights $-2, 0, 2$ on $\omega(\mathrm{IC}_X)$ with multiplicities $2, 3, 2$, respectively.

Proof. For $\alpha \in \Omega$, perverse generic vanishing [13, Theorem 1.1] gives

$$H^k(S, j^*\mathcal{L}_\alpha) = 0 \qquad (k \neq 2).$$

For a unitary rank-one local system, Hodge theory for the associated flat unitary line bundle gives the twisted decomposition

$$H^{p,q}(S, j^*\mathcal{L}_\alpha) = H^q(S, \Omega_S^p \otimes j^*\alpha), \qquad p + q = 2.$$

Equation (3.15), together with Serre duality, gives outer Hodge numbers 2 and 2. Since the Euler characteristic of a rank-one local system equals $e(S) = 8$ and all cohomology is concentrated in degree two, the middle Hodge number is 4. Thus the Hodge vector of $Rj_*\mathbf{Q}_S^H[2]$ is $(2, 4, 2)$. The Hodge-module refinement of (3.6), extended to P, is

$$Rj_*\mathbf{Q}_S^H[2] \simeq \mathrm{IC}_X^H \oplus \mathbf{Q}_0^H(-1).$$

The point summand has type $(1, 1)$, so the Hodge vector for IC_X^H is $(2, 3, 2)$. □

The symmetry of X also gives the self-duality needed in the tensor argument. Indeed, Verdier self-duality of the intersection complex and (3.4) give

$$K^\vee = (-1_P)^* D(K) \simeq K. \tag{3.16}$$

This is the self-duality used below; the same pairing reappears in Section 7.

Lemma 3.8 (The chosen twist is tensor-compatible)**.** *For every $\alpha \in \Omega$, the functor*

$$\omega_\alpha(Q) = H^0(P, Q \otimes \mathcal{L}_\alpha)$$

on the semisimple tensor category generated by the image of $K = \mathrm{IC}_X$ in $\overline{\mathrm{Perv}}(P)$ is a fibre functor. In particular the value $\omega_\alpha(K)$ is the underlying vector space of the twisted Hodge decomposition in Proposition 3.7.

Proof. The decomposition $Rj_*\mathbf{Q}_S[2] \simeq K \oplus \delta_0$ and the definition of Ω give

$$H^i(P, K \otimes \mathcal{L}_\alpha) = 0 \qquad (i \neq 0).$$

Let $m_n : P^n \to P$ be addition. Since a rank-one character local system satisfies $m_n^*\mathcal{L}_\alpha \simeq \mathcal{L}_\alpha^{\boxtimes n}$, the projection formula and Künneth give

$$H^\bullet(P, K^{*n} \otimes \mathcal{L}_\alpha) \simeq H^\bullet(P, K \otimes \mathcal{L}_\alpha)^{\otimes n}, \tag{3.17}$$

so the left-hand side is concentrated in degree zero.

The complexes K^{*n} are projective direct images of pure polarizable Hodge modules, hence the decomposition theorem splits each of their perverse cohomology sheaves as direct summands. Consequently ${}^pH^0(K^{*n})$, and every simple direct summand of it, has twisted cohomology concentrated in degree zero. All other perverse-cohomological pieces are negligible after passage to $\overline{\mathrm{Perv}}(P)$, because convolution is t-exact in the quotient. Moreover, any negligible simple summand of ${}^pH^0(K^{*n})$ has Euler characteristic zero; the preceding concentration then forces its twisted H^0 to vanish. Hence $H^0(- \otimes \mathcal{L}_\alpha)$ factors through the quotient on the tensor category under consideration. Since $K^\vee \simeq K$, every object in the semisimple

tensor category generated by K is a direct summand of a finite direct sum of such ${}^{p}H^{0}(K^{*n})$. Thus the same vanishing holds for every object of that category.

Equation (3.17), by naturality and passage to direct summands, supplies the tensor constraint for ω_α; exactness and faithfulness follow from semisimplicity and the equality $\dim H^0(P, Q \otimes \mathcal{L}_\alpha) = \chi(Q) > 0$ for every nonzero simple object in the quotient. This is the generic-twist fibre-functor construction of [12, Section 3.1, Proposition 3.1]. □

4. The generic pair-incidence splitting

Return to the family over $\mathcal{B}$. Let $\mathcal{E} \to \mathcal{B}$ be the Weierstrass elliptic scheme with zero section O, put $\eta = \mathcal{O}_{\mathcal{E}}(O)$, and let

$$\pi : \mathcal{C} \longrightarrow \mathcal{E}, \qquad \kappa = \pi^{*}\eta$$

be the relative shifted cyclic cover and its theta characteristic. Let $\mathcal{P} \to \mathcal{B}$ be the relative Prym scheme and $\mathcal{X} \subset \mathcal{P}$ the relative centred theta section. For any $Y \to \mathcal{B}$, undecorated symbols E_Y, C_Y, P_Y, X_Y, η and κ denote the corresponding base changes.

The relative theta divisor determined by κ restricts to a section of a relatively ample line bundle on the abelian scheme $\mathcal{P} \to \mathcal{B}$. Its restriction to every closed fibre is the nonzero Cartier divisor constructed in Proposition 3.2. Hence its zero scheme $\mathcal{X} \subset \mathcal{P}$ is a symmetric relative effective Cartier divisor; the defining section is a nonzerodivisor on every fibre of the smooth morphism $\mathcal{P} \to \mathcal{B}$, so $\mathcal{X} \to \mathcal{B}$ is flat.

We next verify these assertions over every geometric point of $\mathcal{B}$. The locus of $b \in \mathcal{B}$ for which X_b is geometrically integral is open for this proper flat family [7, IV$_3$, Théorème 12.2.4(viii)]; it contains every closed point by Proposition 3.2, hence equals $\mathcal{B}$ because $\mathcal{B}$ is Jacobson. Likewise, the nonsmooth locus of $(\mathcal{X} \setminus 0) \to \mathcal{B}$ is closed; if nonempty, it would contain a closed point. The inverse image of the relative smooth locus along the zero section is open in $\mathcal{B}$ and has no closed point, so it is empty. Thus every geometric fibre is integral with singular locus exactly the origin, and the relative smooth locus is

$$\mathcal{M} = (\mathcal{X}/\mathcal{B})_{\mathrm{sm}} = \mathcal{X} \setminus 0.$$

The morphism $\mathcal{M} \to \mathcal{B}$ is smooth and surjective with geometrically integral fibres. Since $\mathcal{B}$ is integral, $\mathcal{M}$ is integral.

Let $\mathcal{N}$ be the relative norm fibre in $\mathcal{C}^{(3)}$. Every Abel class on $\mathcal{M}$ has $h^0 = 1$. On this locus, after choosing a relative Poincaré bundle *étale*-locally, cohomology and base change make its pushforward locally free of rank one, and the relative Abel map is the projectivization of that line bundle. These local isomorphisms descend and are compatible with arbitrary base change. Hence $\mathcal{N} \times_{\mathcal{X}} \mathcal{M} \simeq \mathcal{M}$, and the universal divisor on the relative symmetric cube gives a universal effective divisor of degree three over $\mathcal{M}$.

4.1. **A dominant normalized chart.**

Lemma 4.1 (Dominant normalized chart)**.** *There is a nonempty open subscheme*

$$T_{\mathbf{Q}} \subset \operatorname{Spec} \mathbf{Q}[c, c^{-1}, m, m^{-1}, e_1, e_2, e_3],$$

hence an integral five-dimensional rational scheme, such that with $\lambda = m^{-3}$, after base change to $\mathbf{C}$ suitable dense opens of

$$T = T_{\mathbf{Q}} \times_{\mathbf{Q}} \mathbf{C} \qquad \text{and} \qquad \mathcal{M}$$

are isomorphic. On these opens the universal divisor is described by (4.2) *below. On $e_1 \neq 0$, put*

$$\delta = \frac{c}{e_1^3}, \qquad \Lambda = e_1^3\lambda, \qquad E_2 = \frac{e_2}{e_1^2}, \qquad E_3 = \frac{e_3}{e_1^3}, \qquad Z = e_1 z. \tag{4.1}$$

The residual-octic family and its factor-ordering cover are defined over $T_{\mathbf{Q}}$ and are birationally pulled back from the normalized chart $e_1 = 1$.

Proof. Let D be the unique reduced degree-three divisor representing a point of a dense open in $\mathcal{M}$. Its norm on E is a unique nonvertical line

$$y = mx + n, \qquad m \neq 0.$$

Set

$$x' = mx + n - c, \qquad Y' = y - c, \qquad w' = w.$$

Then the norm line is $Y' = x'$. If the three selected lifts have w'-coordinates α, β, γ, let e_i be their elementary symmetric functions and write

$$p(W) = W^3 - e_1W^2 + e_2W - e_3.$$

Newton's identities give

$$\alpha^3 + \beta^3 + \gamma^3 = e_1^3 - 3e_1e_2 + 3e_3 =: \Sigma_1,$$
$$(\alpha\beta)^3 + (\alpha\gamma)^3 + (\beta\gamma)^3 = e_2^3 - 3e_1e_2e_3 + 3e_3^2 =: \Sigma_2.$$

Consequently

$$\prod_{\xi \in \{\alpha,\beta,\gamma\}} (x' - \xi^3) = (x')^3 - \Sigma_1(x')^2 + \Sigma_2 x' - e_3^3.$$

Since $\lambda = m^{-3}$, the curve and the selected divisor are reconstructed in the primed coordinates by

$$\begin{aligned} (Y' + c)^2 &= G(x') \\ &:= (x' + c)^2 + \lambda\big((x')^3 - \Sigma_1(x')^2 + \Sigma_2 x' - e_3^3\big), \\ D: \quad & x' = Y' = (w')^3, \qquad p(w') = 0. \end{aligned} \tag{4.2}$$

Conversely these formulas recover the original short-Weierstrass model. Indeed write

$$g_2 = 1 - \lambda\Sigma_1, \qquad g_1 = 2c + \lambda\Sigma_2, \qquad g_0 = c^2 - \lambda e_3^3, \qquad n_0 = -\frac{g_2}{3\lambda}.$$

Since $m^3\lambda = 1$, the substitution $x' = mx + n_0$ gives

$$(Y' + c)^2 = x^3 + Ax + B,$$

where

$$A = m\left(g_1 - \frac{g_2^2}{3\lambda}\right), \qquad B = G(n_0).$$

The original norm line is then $y = mx + n_0 + c$. These constructions are inverse on dense opens. They are defined over $\mathbf{Q}$, and therefore construct $T_{\mathbf{Q}}$ with

$$\mathbf{C}(T) = \mathbf{C}(\mathcal{M}). \tag{4.3}$$

The equations are equivariant for

$$(x', Y', w', c) \longmapsto (r^3x', r^3Y', rw', r^3c),$$

which in the original coordinates is the usual Weierstrass scaling

$$(x, y, w, c, A, B) \longmapsto (r^2x, r^3y, rw, r^3c, r^4A, r^6B).$$

Thus

$$(c, \lambda, e_1, e_2, e_3, m) \longmapsto (r^3c, r^{-3}\lambda, re_1, r^2e_2, r^3e_3, rm).$$

The basis of $H^0(C, 3\kappa - D)$ used below has weights $3, 3, 4$, so its projective coordinates transform by

$$[u : v : z] \longmapsto [u : v : r^{-1}z]. \tag{4.4}$$

Taking $r = e_1^{-1}$ gives (4.1) and

$$\mathbf{Q}(c, \lambda, e_1, e_2, e_3) = \mathbf{Q}(\delta, \Lambda, E_2, E_3)(e_1).$$

If $q_0 = m/e_1$, then

$$\mathbf{Q}(T_{\mathbf{Q}}) = \mathbf{Q}(\delta, \Lambda, E_2, E_3)(q_0, e_1), \qquad q_0^3 = \Lambda^{-1}. \tag{4.5}$$

This gives the normalized description used below. □

4.2. **The norm conic.** All formulas in this subsection are first established over the rational function fields indicated below. Put $Y = y - c = w^3$. Since $\deg(3\kappa) = 9 > 2g(C) - 2$, Riemann–Roch gives $h^0(C, 3\kappa) = 6$. For the evaluation map

$$H^0(C, 3\kappa) \longrightarrow H^0(D, (3\kappa)|_D),$$

the kernel has the following basis:

$$s_0 = x - Y, \qquad s_1 = Y - e_1w^2 + e_2w - e_3, \qquad s_2 = (x - e_3)w - e_1Y + e_2w^2.$$

For later use in the norm calculation, the basis argument is included. On D one has $x = Y = w^3$ and $p(w) = w^3 - e_1w^2 + e_2w - e_3 = 0$, so all three sections vanish on D. Moreover $\deg(3\kappa - D) = 6$ and $K_C = 2\kappa$, hence Riemann–Roch gives

$$h^0(C, 3\kappa - D) - h^0(C, D - \kappa) = 3.$$

On the chart under consideration $[\mathcal{O}_C(D) \otimes \kappa^{-1}] \in X_{\mathrm{reg}} = X \setminus \{0\}$, so the degree-zero line bundle $\mathcal{O}_C(D-\kappa)$ is nontrivial and has no section. Thus $h^0(C, 3\kappa-D) = 3$. Finally, in the expansion over $k(E)$ with basis $1, w, w^2$, the coefficient of x in the w-part first forces the coefficient of s_2 in a constant linear relation to vanish; the coefficient of x in the 1-part then forces that of s_0 to vanish, and the coefficient of s_1 follows. The three displayed sections are therefore a basis.

For $s = us_0 + vs_1 + zs_2$, write

$$s = \alpha + \beta w + \gamma w^2,$$

where

$$\begin{aligned} \alpha &= u(x - Y) + v(Y - e_3) - e_1zY, \\ \beta &= e_2v + z(x - e_3), \\ \gamma &= -e_1v + e_2z. \end{aligned}$$

Since $w^3 = Y$, the cyclic norm is

$$\mathrm{Nm}(s) = \alpha^3 + \beta^3Y + \gamma^3Y^2 - 3\alpha\beta\gamma Y. \tag{4.6}$$

It is divisible on E by the fixed norm line $Y - x$. The quotient is a section of $\mathcal{O}_E(6O)$ and hence has a unique expression

$$Q_s = q_{00} + q_{01}x + q_{02}Y + q_{11}x^2 + q_{12}xY + q_{22}Y^2.$$

Let

$$M_s = \begin{pmatrix} q_{00} & q_{01}/2 & q_{02}/2 \\ q_{01}/2 & q_{11} & q_{12}/2 \\ q_{02}/2 & q_{12}/2 & q_{22} \end{pmatrix}.$$

Proposition 4.2 (Determinant factorization)**.** *Put*

$$K_0 = \mathbf{Q}(c, \lambda, e_1, e_2, e_3).$$

In $K_0[u, v, z]$ *one has*

$$\det(M_s) = (u - v)F_8, \tag{4.7}$$

where F_8 *is homogeneous of degree eight and* $u - v \nmid F_8$.

Proof. Each coefficient q_{ij} is homogeneous of degree three in (u, v, z), so $\det(M_s)$ is homogeneous of degree nine. The structural factor can be seen without expanding the determinant. Put

$$h = x - e_1w^2 + e_2w - e_3.$$

The basis above satisfies

$$s_0 + s_1 = h, \qquad s_2 = wh.$$

Hence on the hyperplane $u = v$ one has

$$s = h(u + zw).$$

The section $h \in H^0(C, 2\kappa)$ vanishes on the fixed divisor D. Write $\operatorname{div}(h) = D + D'$. Then

$$[\mathcal{O}_C(D') \otimes \kappa^{-1}] = -x, \qquad \operatorname{Nm}_\pi \mathcal{O}_C(D') = \eta^3.$$

Thus $\pi_* D'$ is a line section of E, and the norm of h is, up to a nonzero scalar, the product of the fixed norm line $Y - x$ and this second line. Moreover

$$\operatorname{Nm}_\pi(u + zw) = u^3 + z^3Y.$$

After dividing by the fixed line, Q_s is therefore a product of two lines whenever $u = v$. Its symmetric matrix has rank at most two, and $\det(M_s)$ vanishes identically on $u = v$. Thus $u - v$ divides the determinant, and the quotient is homogeneous of degree eight.

It remains to show that this factor is simple. Divisibility of F_8 by the monic linear form $u - v$ is equivalent to the identity $F_8(v, v, z) = 0$ over K_0, so one specialization in characteristic zero at which $F_8(v, v, z)$ is nonzero suffices. Set

$$(c, \lambda, e_1, e_2, e_3) = (1, 1, 1, 0, 2), \qquad (u, v, z) = (1 + t, 1, 1).$$

The normalized curve is

$$Y^2 + 2Y = x^3 - 6x^2 + 14x - 8.$$

The six conic coefficients and the exact division by $Y - x$ are written in Appendix A.1. Substitution into the determinant gives

$$4 \det M_{1+t,1,1} = 2t(24t^3 + 56t^2 + 33t + 6). \tag{4.8}$$

Hence $4F_8(1, 1, 1) = 12 \neq 0$ in $\mathbf{Q}$, so $u - v \nmid F_8$. □

Lemma 4.3 (Geometric integrality of the residual octic)**.** *The projective curve*

$$\mathscr{R} = V(F_8) \subset \mathbf{P}^2_{K_0}$$

is geometrically integral.

Proof. Reduce the normalized specialization

$$(c, \lambda, e_1, e_2, e_3) = (1, 1, 1, 0, 2)$$

modulo 13. The corresponding short-Weierstrass curve is

$$y^2 = x^3 + 2x + 5,$$

whose discriminant is nonzero; the branch cubic obtained from $y = 1$ is also separable. Multiplying F_8 by the nonzero scalar 4, set

$$H(u, v) = 4F_8(u, v, 1)$$

at this specialization. It has total degree eight and v-degree eight, with constant leading coefficient 7. At $u = 4$ its monic slice is

$$h(v) = v^8 - 3v^7 - 2v^6 + 3v^5 + 4v^4 + 4v^3 - 3v^2 - 5v - 2 \in \mathbf{F}_{13}[v].$$

Appendix A.2 records the Frobenius reductions

$$v^{13^8} \equiv v \pmod{h}, \qquad \gcd(h, v^{13^4} - v) = 1,$$

together with an explicit Bézout identity for the second relation. The standard factor-degree criterion then makes h irreducible over $\mathbf{F}_{13}$. If H factored in $\mathbf{F}_{13}[u, v]$, the constant leading coefficient in v would force the leading v-coefficients of the factors to be units. A positive-v-degree factor would retain its degree after $u = 4$, while a v-degree-zero factor would itself be a unit. Both possibilities contradict the irreducibility of h. Hence H is irreducible.

Let $F_{8,\mathrm{sp}}(u, v, z)$ denote the homogeneous specialization of F_8. Its dehomogenization at $z = 1$ is $H/4$, and the nonzero v^8 term shows that $z \nmid F_{8,\mathrm{sp}}$. If $F_{8,\mathrm{sp}} = AB$ were a nontrivial homogeneous factorization, then neither $A(u, v, 1)$ nor $B(u, v, 1)$ could be a unit: a positive-degree homogeneous polynomial whose dehomogenization is constant is a scalar multiple of a power of z, which would force z to divide $F_{8,\mathrm{sp}}$. Setting $z = 1$ would therefore give a nontrivial factorization of H, a contradiction. Thus the projective special fibre is irreducible.

The affine curve $H = 0$ contains the rational point $(u, v) = (1, 6)$, where

$$(\partial H/\partial u, \partial H/\partial v) = (2, 0).$$

Thus the point is smooth. Since $\mathbf{F}_{13}$ is perfect, the irreducible special fibre is geometrically reduced. If it were not geometrically irreducible, Frobenius would permute its geometric components transitively. A rational point would then lie on every component and could not be a smooth point. The special fibre is therefore geometrically integral.

Spread the coefficients over a localization $R_0 = \mathbf{Z}[1/N]$ with $13 \nmid N$, and choose an affine open

$$\mathscr{U} \subset \mathbb{A}^5_{R_0} = \operatorname{Spec} R_0[c, \lambda, e_1, e_2, e_3]$$

containing the displayed characteristic-13 point on which the required denominators and one coefficient of F_8 are units. After multiplying by a unit, F_8 defines a flat projective relative Cartier divisor of degree eight in $\mathbf{P}^2_{\mathscr{U}}$. Geometric integrality is

open in such a family [7, IV$_3$, Théorème 12.2.4(viii)]. Since the displayed special fibre is geometrically integral, so is the generic fibre $\mathscr{R}$. □

Lemma 4.4 (Normalized and full function fields)**.** *Work over $e_1 \neq 0$ and put*

$$K_{\mathrm{n}} = \mathbf{Q}(\delta, \Lambda, E_2, E_3).$$

On the affine section chart $v \neq 0$, put

$$U = u/v, \qquad z_0 = z/v, \qquad \zeta = e_1 z_0.$$

Let $\mathscr{R}_{\mathrm{n}}$ be obtained by setting $e_1 = v = 1$ and using coordinates $(\delta, \Lambda, E_2, E_3; U, \zeta)$. Under the parameter map of Lemma 4.1, let

$$\mathscr{R}_{T_{\mathbf{Q}}} = \mathscr{R} \times_{\operatorname{Spec} K_0} \operatorname{Spec} \mathbf{Q}(T_{\mathbf{Q}}).$$

Then

$$k(\mathscr{R}) = k(\mathscr{R}_{\mathrm{n}})(e_1), \qquad k(\mathscr{R}_{T_{\mathbf{Q}}}) = k(\mathscr{R}_{\mathrm{n}})(e_1, q_0), \quad q_0^3 = \Lambda^{-1}. \tag{4.9}$$

The curve $\mathscr{R}_{\mathrm{n}}$ is geometrically integral. After base change to $\mathbf{C}$, the factor-ordering cover on the full chart is the pullback of the normalized one through these field extensions.

Proof. The scaling in Lemma 4.1, together with (4.4), identifies the locus $e_1 v \neq 0$ birationally with $\mathbf{G}_m \times \mathscr{R}_{\mathrm{n}}$. This gives the first equality in (4.9). The chart $v \neq 0$ is dense because $\mathscr{R}$ is a geometrically integral octic. Geometric integrality of $\mathscr{R}_{\mathrm{n}}$ follows by faithfully flat descent from Lemma 4.3. The second equality is (4.5). Formation of the two line factors of a rank-two conic commutes with field extension, which proves the last assertion. □

Lemma 4.5 (Off-curve norm identity)**.** *With*

$$P_D(x) = x^3 - \Sigma_1 x^2 + \Sigma_2 x - e_3^3$$

and

$$F_E(x, Y) = (Y + c)^2 - (x + c)^2 - \lambda P_D(x),$$

the cyclic norm satisfies the polynomial identity

$$\operatorname{Nm}(s) = (Y - x) Q_s - \frac{v^3 + z^3 Y}{\lambda} F_E(x, Y). \tag{4.10}$$

In particular, no coefficient identity for the residual octic is needed to recover the two residual line factors.

Proof. Put

$$R = \lambda \operatorname{Nm}(s) + (v^3 + z^3 Y) F_E(x, Y).$$

The only term of ordinary degree four in $\lambda \operatorname{Nm}(s)$ is $\lambda z^3 x^3 Y$, coming from $\lambda \beta^3 Y$; it is cancelled by the term $-\lambda z^3 x^3 Y$ in $(v^3 + z^3 Y) F_E$. Hence R has ordinary degree at most three in x, Y.

On the line $Y = x$ one has $s_0 = 0$, $s_1 = h$, and $s_2 = wh$, where

$$h = x - e_1 w^2 + e_2 w - e_3.$$

Since $w^3 = x$ on that line, $h = p(w)$ and

$$\operatorname{Nm}(h) = \operatorname{Res}_W(W^3 - x, p(W)) = P_D(x).$$

It follows that

$$\operatorname{Nm}(s)|_{Y=x} = (v^3 + z^3 x) P_D(x), \qquad F_E(x, x) = -\lambda P_D(x).$$

Thus $R|_{Y=x} = 0$, so $Y - x$ divides R. The quotient $R/(\lambda(Y - x))$ has degree at most two and, on $E = V(F_E)$, restricts to $\mathrm{Nm}(s)/(Y-x)$. By the uniqueness of the six-term expression defining Q_s, this quotient is Q_s. Equation (4.10) follows. □

Proposition 4.6 (Cubic-norm splitting of the ordering cover)**.** *After base change to* $\mathbf{C}$*, there is a dense normal open* $\mathscr{R}^\circ_{\mathbf{C}}$ *of the rank-two locus of the residual octic such that the intrinsic factor-ordering cover is split:*

$$\mathrm{Ord}_{\mathrm{fac},\mathscr{R}^\circ_{\mathbf{C}}} \simeq \mathscr{R}^\circ_{\mathbf{C}} \sqcup \mathscr{R}^\circ_{\mathbf{C}}. \tag{4.11}$$

The same conclusion holds after pullback to a dense normal open of $\mathscr{R}_T$*.*

Proof. Let $L = k(\mathscr{R}_{\mathbf{C}})$ and let $[u : v : z]$ be the generic section parameter. Geometric integrality of $\mathscr{R}$ allows restriction to the dense chart

$$vz(u - v) \neq 0.$$

The generic residual conic has rank two. Indeed, if its symmetric matrix had rank at most one at the generic point, then its adjugate would vanish. The differential formula

$$d(\det M_s) = \mathrm{tr}(\mathrm{adj}(M_s)\, dM_s)$$

would then make every first derivative of $\det M_s$ vanish there. Since

$$\det M_s = (u - v)F_8$$

and $u - v \neq 0$ in L, all first derivatives of F_8 would vanish at the generic point. This is impossible for the geometrically integral plane curve $F_8 = 0$ in characteristic zero.

Put

$$q = -\frac{v}{z}, \qquad Y_* = q^3,$$

and choose a primitive cube root of unity $\omega \in \mathbf{C}$. On the line $Y = Y_*$ the cubic algebra $w^3 = Y_*$ splits, and

$$\mathrm{Nm}(s)(x, Y_*) = \prod_{j=0}^{2} L_j(x), \qquad L_j(x) = s(x, \omega^j q) \in L[x].$$

The identities $s_0 + s_1 = h$ and $s_2 = wh$ give

$$s = (u + zw)s_0 + (v + zw)s_1.$$

For $w = q$ the second coefficient vanishes, so

$$L_0(x) = (u - v)(x - Y_*).$$

At $Y = Y_*$ the correction term in (4.10) vanishes. Cancelling $x - Y_*$ therefore gives

$$Q_s(x, Y_*) = -(u - v)L_1(x)L_2(x). \tag{4.12}$$

Write $L_j = a_j x + b_j$ for $j = 1, 2$. Since $q = -v/z$,

$$a_j = u - v\omega^j,$$

and neither a_1 nor a_2 vanishes at the generic point of the irreducible degree-eight curve $\mathscr{R}_{\mathbf{C}}$. Shrink once so that $a_1 a_2 \neq 0$. Then

$$b_j = -(u - v\omega^j)q^3 + v(1 - \omega^j)\big(q^3 - e_1\omega^{2j}q^2 + e_2\omega^j q - e_3\big).$$

Using only $\omega^2 + \omega + 1 = 0$, these two linear forms give

$$a_1 b_2 - a_2 b_1 = -\frac{v(2\omega + 1)}{z^3}\, R(u, v, z), \tag{4.13}$$

where

$$R = e_1uv^2z + 2e_1v^3z - 2e_2uvz^2 - e_2v^2z^2 + e_3uz^3 - e_3vz^3 + uv^3 - v^4.$$

The polynomial R is nonzero and homogeneous of degree four in (u, v, z). Since F_8 is irreducible of degree eight, R does not vanish at the generic point of $\mathscr{R}_{\mathbf{C}}$. After one further shrinking, the two affine roots in (4.12) are therefore distinct.

A rank-two conic has a unique singular point, namely the projectivized one-dimensional kernel of M_s; this point is L-rational. Each of the two factor lines is determined by that singular point and its intersection with the L-rational line $Y = Y_*$. Equation (4.12), together with (4.13), shows that both intersection points are L-rational and distinct. Hence both factor lines are individually defined over L.

The factor-ordering cover thus has two rational sections over the generic point. Because it is finite *étale* on the rank-two locus, the two sections extend after shrinking to a dense normal open, proving (4.11). Pullback along $\mathscr{R}_T \to \mathscr{R}$ preserves this splitting. □

4.3. **The scheme-theoretic Abel inverse.** Over the chart T of Lemma 4.1, let $p : C_T \to T$ be the relative curve and let D be its universal divisor. After restricting T, the sheaf $p_*\mathcal{O}_{C_T}(3\kappa - D)$ is locally free of rank three. The residual equation $F_8 = 0$ defines a relative octic

$$\mathscr{R}_T \subset \mathbf{P}_T\big(p_*\mathcal{O}_{C_T}(3\kappa - D)\big).$$

After a further shrinking of T, the section F_8 is nonzero on every projective-space fibre. Thus $\mathscr{R}_T \to T$ is a flat projective relative Cartier divisor. Its generic fibre is geometrically integral by Lemmas 4.3 and 4.4. Flatness forces every irreducible component to dominate the integral base T, so the integral generic fibre gives a single component. A nilpotent section would vanish generically and hence be $\mathcal{O}_T$-torsion, which flatness excludes. Thus $\mathscr{R}_T$ is integral.

Let $\mathscr{R}_T^{(2)} \subset \mathscr{R}_T$ be the open subset on which the residual conic has rank exactly two. The relative Hilbert scheme of line components of the universal conic is then finite étale of degree two over $\mathscr{R}_T^{(2)}$. Choosing one component determines the other and hence orders them; denote this cover by

$$\mathrm{Ord}_{\mathrm{fac}} \longrightarrow \mathscr{R}_T^{(2)}.$$

Its points will be written $([s], \ell_1, \ell_2)$, with $Q_s = \ell_1\ell_2$ up to a unit pulled back from the base.

Definition 4.7 (Admissible residual-norm locus)**.** Let $\mathscr{G} \subset \mathscr{R}_T^{(2)}$ be the open locus on which the following conditions hold after pullback to $\mathrm{Ord}_{\mathrm{fac}}$:

(i) $u - v \neq 0$, excluding the structural determinant component;
(ii) the fixed divisor D is reduced;
(iii) the fixed norm line and the two lines $\ell_i = 0$ meet E transversely, their degree-three line sections are pairwise disjoint, and the residual line sections avoid the branch divisor of $C \to E$;
(iv) above each residual point, s vanishes on exactly one sheet of $C \to E$, and every such zero is simple;

(v) the two resulting degree-three zero divisors have $h^0 = 1$, and their Abel classes lie in the relative smooth locus of $\mathcal{X}$.

These conditions define an open subset of $\mathrm{Ord}_{\mathrm{fac}}$ invariant under interchanging ℓ_1 and ℓ_2. Its image is open because the finite étale cover is open, and invariance makes it the full inverse image of that image. This defines $\mathscr{G}$.

Lemma 4.8 (Relative ordered-zero construction)**.** *Put*

$$\mathrm{Ord}_{\mathrm{fac},\mathscr{G}} = \mathrm{Ord}_{\mathrm{fac}} \times_{\mathscr{R}_T^{(2)}} \mathscr{G}.$$

Let $\mathrm{Ord}_{\mathrm{div}}$ *be the functor over* $\mathscr{G}$ *whose* W*-points are ordered pairs of relative effective Cartier divisors* (D_1, D_2) *of degree three on* C_W/W *such that*

$$\mathrm{div}(s_W) = D_W + D_1 + D_2$$

and the unordered pair $\{\pi_* D_1, \pi_* D_2\}$ *is the unordered pair of line sections of* E_W *defined by the two components of* Q_{s_W}*. This functor is represented by a finite étale double cover of* $\mathscr{G}$*. The assignment* $Q_s = \ell_1\ell_2 \mapsto (D_1, D_2)$ *is an isomorphism of finite étale double covers*

$$\mathrm{Ord}_{\mathrm{fac},\mathscr{G}} \xrightarrow{\sim} \mathrm{Ord}_{\mathrm{div}}\,.$$

It commutes with arbitrary base change, including nonreduced base change. Moreover, each D_i *is a relative effective Cartier divisor of degree three, and* $[\mathcal{O}(D_i) \otimes \kappa^{-1}]$ *lies in* $\mathcal{X}_{\mathrm{sm}/\mathcal{B}}$*.*

Proof. For the forward construction put $T' = \mathrm{Ord}_{\mathrm{fac},\mathscr{G}}$ and write $Q_s = \ell_1\ell_2$ in its universal ordering. By definition of $\mathscr{G}$, none of the tangency, branch, collision, multiple-zero, or Abel speciality loci excluded in Definition 4.7 occurs. For the remainder of the forward construction, every object with a subscript T is pulled back to T'; this base change is suppressed from the notation. The line ℓ_i then cuts out a relative finite étale divisor $B_i \subset E_T$ of degree three. Moreover

$$U_i = C_T \times_{E_T} B_i \longrightarrow B_i$$

is finite étale of degree three, because B_i avoids the branch divisor.

The zero over B_i is selected scheme-theoretically as follows. Étale-locally on B_i, split $U_i = B_i \sqcup B_i \sqcup B_i$ and trivialize the restriction of 3κ, using the induced trivialization of its norm line. If $a_1, a_2, a_3 \in \mathcal{O}_{B_i}$ are the three values of s, compatibility of the finite locally free norm with base change gives

$$a_1 a_2 a_3 = \mathrm{Nm}_\pi(s)|_{B_i} = 0 \tag{4.14}$$

as an equality in the structure sheaf, not merely at geometric points. The collision exclusions imply that locally two values, say a_2, a_3, are nonzero on every geometric fibre. Their vanishing loci are therefore empty, so they are units in the corresponding structure sheaf. Equation (4.14) then forces $a_1 = 0$, also in nilpotent directions. Consequently

$$Z_i = Z(s) \times_{E_T} B_i \subset U_i$$

is locally exactly one open-and-closed sheet of U_i/B_i. Thus $Z_i \to B_i$ is finite étale of degree one and hence is an isomorphism.

Let t be a local equation for B_i in E_T. Since B_i occurs with multiplicity one in the norm divisor and the other norm factors are disjoint from it, $\mathrm{Nm}_\pi(s) = tu$ with u a unit along B_i. Étale-locally on a neighbourhood of B_i in the complement of the branch divisor, split the cover $C_T \to E_T$. On the selected sheet the other two values of s remain units, so the same local product formula shows that the selected

value is t times a unit. The zero is therefore simple. The image $D_i \subset C_T$ of Z_i is finite étale of degree three over T'; as a finite étale multisection of the smooth relative curve C_T/T', it is a relative effective Cartier divisor.

The pairwise disjoint divisors D, D_1, D_2 lie in $Z(s)$ and have total relative degree nine, equal to $\deg(3\kappa)$. After dividing by their Cartier equations, the residual section has degree zero and is nonvanishing on every geometric fibre. Nakayama's lemma makes it a trivialization, so there is no vertical or nilpotent residual contribution and

$$Z(s) = D + D_1 + D_2 \tag{4.15}$$

as relative Cartier divisors. Conversely, an ordered divisor datum determines the unique projective section $[s]$ with zero divisor $D + D_1 + D_2$, and taking its norm recovers the ordered line sections B_i and hence the ordered factors ℓ_i. Thus these assignments are inverse already at the divisor level.

It remains to compare this divisor construction with the relative Abel incidence. Let

$$\alpha : C_T^{(3)} \longrightarrow \mathrm{Pic}^3(C_T/T)$$

be the relative Abel map and let $\mathcal{F} = \mathrm{Nm}^{-1}(\eta^3) \subset \mathrm{Pic}^3(C_T/T)$. The norm fibre is the cartesian base change

$$N_T = C_T^{(3)} \times_{\mathrm{Pic}^3(C_T/T)} \mathcal{F}.$$

Let W_3^0 be the determinantal theta subscheme of $\mathrm{Pic}^3(C_T/T)$ and $W_3^1 \subset W_3^0$ its locus $h^0 \geq 2$. Then $W_3^{(1)} = W_3^0 \setminus W_3^1$ is open in theta. Locally, the derived pushforward of a Poincaré bundle is represented by a map of equal-rank vector bundles, since the degree-three Euler characteristic on a genus-four curve is zero. At a point with $h^0 = 1$, an invertible minor reduces this map to $\mathrm{diag}(I_{n-1}, f)$; the theta ideal is (f) [11]. On $W_3^{(1)}$ its kernel and cokernel are thus line bundles and commute with arbitrary base change, including nonreduced base change. The Abel source is the projectivization of the kernel, so

$$\alpha^{-1}(W_3^{(1)}) \xrightarrow{\sim} W_3^{(1)}.$$

Base changing this isomorphism by $\mathcal{F} \to \mathrm{Pic}^3(C_T/T)$ and translating by κ^{-1} gives

$$N_T^{(1)} \xrightarrow{\sim} X_T^{(1)}.$$

Here, by definition, $X_T^{(1)}$ is the open part of the theta section represented by line bundles with $h^0 = 1$. Restrict both sides to the inverse images of $X_T^{(1)} \cap X_{T,\mathrm{sm}/T}$, as imposed above. Thus the divisor construction descends to the relative smooth Abel incidence.

Zero schemes, fibre products, finite locally free norms, and the cartesian Abel construction all commute with arbitrary base change. Hence both composites preserve the universal zero schemes and are the identity as morphisms. The use of $C_T^{(3)} \simeq \mathrm{Hilb}^3(C_T/T)$ is only representability; no normalization of the ordering cover or of pair fibres is involved. □

Let $x_T : T \to \mathcal{M}$ be the point represented by the universal divisor D, so that $x_T = [\mathcal{O}(D) \otimes \kappa^{-1}]$. Put

$$h = x - e_1 w^2 + e_2 w - e_3.$$

In the basis used for (4.7), direct substitution gives

$$s_0 + s_1 = h, \qquad s_2 = wh. \tag{4.16}$$

The divisor of h is $D + D_{-x_T}$: it contains D, and its residual degree-three divisor has Abel class $-x_T$ and is unique on the $h^0 = 1$ locus.

Define

$$\mathcal{I}_T^\dagger = \{y \in X_T^{(1)} \cap X_{T,\mathrm{sm}/T} : -x_T - y \in X_T^{(1)} \cap X_{T,\mathrm{sm}/T}\}.$$

The two relative Abel inverses give universal divisors D_1, D_2 on $C_T \times_T \mathcal{I}_T^\dagger$. Their Abel classes satisfy

$$[\mathcal{O}(D + D_1 + D_2) \otimes \kappa^{-3}] = 0 \quad \text{in } \mathrm{Pic}^0(C_T/T).$$

Thus $\mathcal{O}(D_1 + D_2)$ and $\mathcal{O}(3\kappa - D)$ differ only by a line bundle pulled back from the base. Their divisor sections determine a unique projective relative section

$$\vartheta : \mathcal{I}_T^\dagger \longrightarrow \mathbf{P}_T\big(p_*\mathcal{O}_{C_T}(3\kappa - D)\big).$$

The structural determinant line requires separate treatment: on $u = v$ the identity (4.16) gives

$$s = h(u + zw), \qquad \mathrm{div}(s) = D + D_{-x_T} + D_\kappa, \quad D_\kappa \in |\kappa|.$$

For a general pencil member the only residual Abel classes are $-x_T$ and 0. Exceptional parameters may admit nontrivial repartitions; consequently the structural-line incidence can be nonempty even though its generic member contributes only the two excluded classes. Set

$$\mathcal{I}_{T,\mathrm{res}}^\dagger = \mathcal{I}_T^\dagger \cap \vartheta^{-1}(u - v \neq 0).$$

Two residual norm line sections imply $\vartheta^*(\det M_s) = 0$ scheme-theoretically. Since $\det M_s = (u-v)F_8$, dividing by the invertible $u - v$ yields $\vartheta^* F_8 = 0$ on this residual open, including nonreduced test schemes. Thus ϑ factors through $\mathscr{R}_T$ on this open. Define

$$\mathcal{I}_T^{\mathrm{adm}} = \mathcal{I}_{T,\mathrm{res}}^\dagger \cap \vartheta^{-1}(\mathscr{G}).$$

Proposition 4.9 (Abel-incidence identification)**.** *There is an isomorphism over $\mathscr{G}$*

$$\Phi : \mathrm{Ord}_{\mathrm{fac},\mathscr{G}} \xrightarrow{\sim} \mathcal{I}_T^{\mathrm{adm}}, \qquad ([s], \ell_1, \ell_2) \longmapsto [\mathcal{O}(D_1) \otimes \kappa^{-1}]. \tag{4.17}$$

Interchanging ℓ_1 and ℓ_2 corresponds to the involution $y \mapsto -x_T - y$. The isomorphism commutes with arbitrary base change.

Proof. On $\mathrm{Ord}_{\mathrm{fac},\mathscr{G}}$, put

$$y_i = [\mathcal{O}(D_i) \otimes \kappa^{-1}] \in P_T.$$

The divisor equality (4.15) gives

$$x_T + y_1 + y_2 = [\mathcal{O}(D + D_1 + D_2) \otimes \kappa^{-3}] = 0. \tag{4.18}$$

Hence $y_2 = -x_T - y_1$, and the ordered-zero construction of Lemma 4.8 defines the morphism Φ. The defining conditions of $\mathscr{G}$ show that it lands in $\mathcal{I}_T^{\mathrm{adm}}$. Conversely, let y be a W-point of $\mathcal{I}_T^{\mathrm{adm}}$, write $p_W : C_T \times_T W \to W$ for the projection, put $y_1 = y$ and $y_2 = -x_T - y$, and let D_1, D_2 be their relative Abel inverse divisors. Reading (4.18) from right to left gives

$$\mathcal{O}(D_1 + D_2) \simeq \mathcal{O}(3\kappa - D) \otimes p_W^*\mathcal{N}$$

for a line bundle $\mathcal{N}$ on W. The divisor section therefore gives a unique projective section $[s]$ with $\operatorname{div}(s) = D + D_1 + D_2$; the factor pulled back from W does not affect this projective point. Since $y_i \in P_T$,

$$\operatorname{Nm}_\pi \mathcal{O}(D_i) = \operatorname{Nm}_\pi(\kappa) = \eta^3.$$

Thus the effective divisor $\pi_* D_i$ on E_T is cut out by a unique projective line ℓ_i. Multiplicativity of the finite locally free norm gives, up to a unit from W,

$$\operatorname{Nm}_\pi(s) = \ell_D \ell_1 \ell_2, \qquad Q_s = \ell_1 \ell_2,$$

where ℓ_D is the fixed norm line of D. Admissibility makes the two residual factors distinct and orders them by the ordered pair (D_1, D_2). This constructs the inverse of (4.17).

The relative Abel inverse on the $h^0 = 1$ locus, formation of zero divisors, and finite locally free norms commute with arbitrary base change. The two composites preserve the universal Cartier divisors and hence are the identity as morphisms. Swapping D_1, D_2 gives $y_1 \leftrightarrow y_2$, which is the stated incidence involution. □

Lemma 4.10 (Nonemptiness of the admissible locus)**.** *The open subscheme $\mathscr{G}$ of Definition 4.7 is nonempty in characteristic zero. It is dense in $\mathscr{R}_T$, dominates T, and its geometric generic fibre is a nonempty dense open of the geometric generic residual curve.*

Proof. For nonemptiness, use the following characteristic-13 point of the normalized chart:

$$(c, m, e_1, e_2, e_3) = (1, 1, 1, 0, 2), \qquad \lambda = m^{-3} = 1, \qquad (u : v : z) = (3 : 5 : 1).$$

Direct substitution verifies all four conditions of Definition 4.7: the fixed divisor is reduced; the three norm lines cut transverse, pairwise disjoint sections away from the branch divisor; each residual point lifts to one simple zero; and both residual divisors lie in the smooth $h^0 = 1$ Abel locus. Every coefficient, discriminant, intersection point, and nonvanishing test in this substitution is displayed in Appendix A.3.

For the passage to characteristic zero, enlarge the localization $R_0 = \mathbf{Z}[1/N]$ from Lemma 4.3, without inverting 13, so that the chart and all admissibility conditions spread out. Work on the affine chart $v \neq 0$. The coordinate ring A_0 of a suitable affine open in the spread of $T_{\mathbf{Q}}$ is a localization of

$$R_0[c, c^{-1}, m, m^{-1}, e_1, e_2, e_3]$$

and hence is a unique factorization domain. With $U_0 = u/v$ and $Z_0 = z/v$, clear denominators in $F_8(U_0, 1, Z_0)$ and divide by its content. The resulting primitive polynomial $f \in A_0[U_0, Z_0]$ is irreducible over $\operatorname{Frac}(A_0)$ by Lemmas 4.3 and 4.4; Gauss's lemma makes it irreducible in $A_0[U_0, Z_0]$. Thus

$$\mathscr{Y} = \operatorname{Spec} A_0[U_0, Z_0]/(f)$$

is integral. Since R_0 injects into this domain, its coordinate ring is torsion-free as an R_0-module; over the localization R_0 of $\mathbf{Z}$ this is equivalent to flatness. Hence $\mathscr{Y}$ is R_0-flat.

Admissibility defines an open subscheme of $\mathscr{Y}$ containing the displayed point over $\mathbf{F}_{13}$. It therefore contains a principal open $D(g)$ with $g \neq 0$. Since $\mathscr{Y}$ is integral, $D(g)$ contains the generic point of $\mathscr{Y}$; in particular, its characteristic-zero generic fibre is nonempty. The generic residual curve dominates the generic point

of $T_{\mathbf{Q}}$, and this remains true on $D(g)$. After base change to $\mathbf{C}$, this open lies in $\mathscr{G}$. Hence $\mathscr{G}$ is nonempty, dense in $\mathscr{R}_T$, and dominates T; its geometric generic fibre is a nonempty open of a geometrically integral curve and is therefore dense. □

Corollary 4.11 (The two admissible branches)**.** *After replacing $\mathscr{G}$ by a dense open subset, there is an isomorphism over $\mathscr{G}$*

$$\mathcal{I}_T^{\mathrm{adm}} \simeq \mathscr{G} \sqcup \mathscr{G}.$$

The incidence involution exchanges the two copies. On a geometric generic fibre over T, each copy is a dense open of a geometrically integral curve.

Proof. By Proposition 4.6, the finite étale factor-ordering cover has two rational sections over $\mathbf{C}$ at the generic point of $\mathscr{G}$. They extend to two disjoint sections after shrinking $\mathscr{G}$, so $\mathrm{Ord}_{\mathrm{fac},\mathscr{G}} \simeq \mathscr{G} \sqcup \mathscr{G}$. Apply Proposition 4.9. The last assertion follows from Lemmas 4.3 and 4.10. □

4.4. **Boundary exhaustion and global labels.** Let ξ be the generic point of the integral scheme $\mathcal{M}$, and let $\bar{\xi}$ be a geometric generic point. For a geometric point x of $\mathcal{M}$, define the pair fibre scheme-theoretically by

$$C_x = X \cap (-x - X) \subset P, \qquad C_x^{\mathrm{ne}} = C_x \setminus \{0, -x\}.$$

Over $\bar{\xi}$, the unique effective divisors representing $x, y, -x-y$ satisfy

$$D_x + D_y + D_{-x-y} = \mathrm{div}(s_y) \tag{4.19}$$

for a projective section $[s_y] \in \mathbf{P}H^0(C, 3\kappa - D_x)$. This defines a rational map

$$\Psi : C_x \dashrightarrow \mathbf{P}H^0(C, 3\kappa - D_x).$$

Lemma 4.12 (Boundary exhaustion)**.** *The rational map Ψ has finite fibres wherever it is defined. The determinant component $u = v$ gives no nonexceptional incidence-curve component; its generic divisor data give only the Abel classes $y = 0, -x$. Every irreducible curve component of $C_{\bar{\xi}}^{\mathrm{ne}}$ dominates the residual octic $\mathscr{R}$, meets its admissible open, and is the closure of one of the two branches in Corollary 4.11. In particular, $C_{\bar{\xi}}^{\mathrm{ne}}$ has exactly two irreducible components.*

Proof. For a fixed section, its effective zero divisor on the smooth curve C has only finitely many effective degree-three subdivisors, so Ψ has finite fibres. Recall h and the identities in (4.16). The zero divisor of $h \in H^0(C, 2\kappa)$ is $D_x + D_{-x}$. Indeed, it contains D_x, and the residual degree-three divisor has Abel class $-x$ and is unique on the generic $h^0 = 1$ locus. Equation (4.16) identifies the line $u = v$ with

$$\mathbf{P}\big(h \cdot H^0(C, \kappa)\big) = \mathbf{P}H^0(C, 3\kappa - D_x - D_{-x}).$$

Its members have divisor $D_x + D_{-x} + D_\kappa$ with $D_\kappa \in |\kappa|$. For a general member of this pencil, $\pi_* D_{-x}$ and $\pi_* D_\kappa$ are reduced, disjoint line sections of the plane cubic E. A degree-three subcycle using two points from the first section and one from the second cannot itself be a line section: a line through the first two points is the first line and cannot contain the third point. The same argument applies with the roles reversed. Hence the only degree-three subcycles whose pushforward is a line section are the two complete triples. Their Abel classes are $-x$ and 0.

Reducedness, transversality, and disjointness fail on proper closed subsets of this projective line, so the exceptional parameters of the structural pencil form a finite set. If an incidence-curve component mapped dominantly to $u = v$, its generic point would contradict the preceding paragraph; if it mapped to one parameter, it would

contradict the finiteness of the fibres of Ψ. Thus $u = v$ gives no nonexceptional curve component.

The same two-line argument shows that, over $\mathscr{G}$ the only Prym degree-three subcycles of the six residual zeros are the two complete line sections. By Proposition 4.9 and Corollary 4.11, these give exactly the two displayed admissible branches.

By the uniqueness of the g^1_3, the only points where an Abel inverse in the definition of Ψ is not unique are $y = 0$ and $y = -x$. Thus the indeterminacy of Ψ is contained in these two exceptional points.

Where Ψ is defined, (4.19) and multiplicativity of the finite locally free norm give, up to a nonzero scalar,

$$\mathrm{Nm}_\pi(s_y) = \ell_{D_x}\ell_{D_y}\ell_{D_{-x-y}}.$$

After division by the fixed line ℓ_{D_x}, the residual conic is the product $Q_{s_y} = \ell_{D_y}\ell_{D_{-x-y}}$. Hence $\operatorname{rank} M_{s_y} \le 2$ and $\det M_{s_y} = 0$. The image of Ψ is therefore contained in

$$V(\det M_s) = V(u - v) \cup V(F_8).$$

The preceding paragraph excludes $V(u-v)$ as the image of a nonexceptional curve component. Consequently every nonexceptional curve component maps to the residual octic $V(F_8)$. Let Z be such a component. Its image is one-dimensional because $\Psi|_Z$ has finite fibres, and hence is dense because $\mathscr{R}$ is geometrically integral. Thus $Z \cap \Psi^{-1}(\mathscr{G})$ is a dense open subset of Z. Over $\mathscr{G}$, Proposition 4.9 and Corollary 4.11 identify the incidence scheme with the disjoint union of two irreducible branches. The irreducibility of Z places its dense open in exactly one of them, and dimensional equality then makes it dense in that branch. Hence Z is the closure of one of the two admissible branches. Conversely, each branch closure is an irreducible component. These are therefore exactly the two nonexceptional components. Here $\mathbf{C}(T) = \mathbf{C}(\mathcal{M})$ from (4.3) identifies the two geometric generic fibres. □

Proposition 4.13 (The geometric generic pair fibre)**.** *The scheme $C_{\bar\xi}$ is reduced, and $C^{\mathrm{ne}}_{\bar\xi}$ has exactly two geometrically integral irreducible components $A_{\bar\xi}$ and $B_{\bar\xi}$. They are exchanged by $y \mapsto -x - y$, and both descend to $k(\xi)$.*

Proof. By Lemmas 4.3, 4.10 and 4.12, Propositions 4.6 and 4.9, and Corollary 4.11, the nonexceptional geometric generic fibre has exactly two geometrically integral components, exchanged by the incidence involution. Since $\mathrm{Stab}_P(X) = \{0\}$ and x is generic, the integral Cartier divisors X and $-x - X$ are distinct. Their local equations form a regular sequence in the smooth threefold P, so $C_{\bar\xi}$ is a pure local complete intersection curve. It is therefore Cohen–Macaulay and has no embedded associated points. The two dense admissible branches show that it is reduced at every generic point. Hence $C_{\bar\xi}$ is reduced.

It remains to descend the two labels. Let τ be the order-three deck transformation. On P one has

$$1 + \tau + \tau^2 = 0,$$

and $\tau(X) = X$. Therefore

$$y_+(x) = \tau x \in C_x, \qquad s_+(x) = (x, \tau x)$$

defines a rational section of the pair-incidence family; the associated ordered pair is $(\tau x, \tau^2 x)$. This section meets the smooth locus of the projection of the pair-incidence family to the x-coordinate on a nonempty open. If it did not, then for

general x the tangent hyperplanes to X at τx and $\tau^2 x$ would coincide, so their Gauss covectors would be equal. The equivariance identity

$$\gamma_X(\tau x) = \tau \cdot \gamma_X(x)$$

then makes that common covector projectively fixed by τ. The general Gauss image would then lie in the projective fixed locus of τ, contrary to the dominance in (3.9). Explicitly,

$$T_0^* P = \left\langle \frac{dx}{w(w^3+c)} \right\rangle \oplus \left\langle \frac{dx}{w^2(w^3+c)}, \frac{x\,dx}{w^2(w^3+c)} \right\rangle,$$

and the two summands have distinct τ-eigenvalues. The projective fixed locus is a point union a line, a proper subset of $\mathbf{P}^2$.

Over $k(\xi)$, the generic value of y_+ is a smooth rational point and lies on exactly one geometric component. Galois therefore fixes the component containing y_+ and, since there are exactly two components, also fixes the other. Both components therefore descend to $k(\xi)$. □

Remark 4.14 (The structural-pencil boundary). At geometric generic x, nonexceptional incidence above the structural pencil may contain isolated points but no curve components. Write $D_{-x} = p_1 + p_2 + p_3$. A moving trigonal divisor $D_\kappa(q)$ is cut out by $w = q$ and has norm the line $y = c + q^3$ on E. Overlap with $\pi_* D_{-x}$ forces $q^3 = w(p_i)^3$ for some i, giving at most nine finite affine parameters; infinity and all further degeneracies form a finite set. Elsewhere the two norm triples are disjoint, and no mixed three-point subcycle can be a line section: two points on one original line already determine that line and exclude a third point on the other. Thus only the two complete triples, of classes $-x$ and 0, remain. The incidence-to-section map has finite fibres, since a fixed degree-nine zero divisor has finitely many degree-three subdivisors. Hence no nonexceptional curve component can dominate $u = v$ or be contracted to an exceptional pencil parameter.

5. Spreading the generic splitting

Define the relative pair-incidence scheme

$$\mathcal{I} = \{(x, y) \in \mathcal{M} \times_{\mathcal{B}} \mathcal{X} : -x - y \in \mathcal{X}\},$$

with the scheme structure induced by the two theta equations, and let $q : \mathcal{I} \to \mathcal{M}$ be the first projection. The equations $y = 0$ and $y = -x$ define two sections of q. Let $\mathcal{I}^{\mathrm{ne}}$ denote their complement.

Theorem 5.1 (Uniform pair splitting)**.** *There are a nonempty open subscheme $\mathcal{M}^\circ \subset \mathcal{M}$ and closed subschemes $\mathcal{A}_1, \mathcal{A}_2 \subset \mathcal{I}|_{\mathcal{M}^\circ}$ with the following properties.*

(i) *Each $\mathcal{A}_i \to \mathcal{M}^\circ$ is flat, and every geometric fibre is a geometrically integral curve.*

(ii) *Every geometric fibre of $q|_{\mathcal{M}^\circ}$ is reduced, and*

$$\mathcal{I}^{\mathrm{ne}}|_{\mathcal{M}^\circ} = (\mathcal{A}_1 \cup \mathcal{A}_2) \cap \mathcal{I}^{\mathrm{ne}}$$

scheme-theoretically.

(iii) *The two fibre curves are distinct at every geometric point of $\mathcal{M}^\circ$, and the involution $(x, y) \mapsto (x, -x - y)$ exchanges $\mathcal{A}_1$ and $\mathcal{A}_2$.*

The image

$$\mathcal{B}^\circ = \mathrm{im}(\mathcal{M}^\circ \to \mathcal{B})$$

is a nonempty Zariski-open subset. For every $b \in \mathcal{B}^\circ(\mathbf{C})$*, the fibre*

$$U_b = \mathcal{M}^\circ \times_{\mathcal{B}} \{b\}$$

is a nonempty dense connected open subset of $(X_b)_{\mathrm{reg}}$*. For every* $x \in U_b(\mathbf{C})$*, the curves* $(\mathcal{A}_1)_x$ *and* $(\mathcal{A}_2)_x$ *are exactly the two nonexceptional irreducible components of* C_x*.*

Proof. Throughout the proof the schemes may be restricted to a smaller nonempty open of $\mathcal{M}$ without changing notation; the final open is denoted $\mathcal{M}^\circ$. By Proposition 4.13, the geometric generic fibre of $q : \mathcal{I} \to \mathcal{M}$ is reduced and has exactly two geometrically integral components, both defined over $k(\mathcal{M})$. Let $\mathcal{A}_1, \mathcal{A}_2 \subset \mathcal{I}$ be their closures. The involution exchanges them because it exchanges their generic fibres. The morphism q is projective and of finite presentation. Apply generic flatness to q and to $\mathcal{A}_i \to \mathcal{M}$. Geometric reducedness of the fibres of q and geometric integrality of the fibres of each $\mathcal{A}_i$ then hold after restricting to a nonempty open of $\mathcal{M}$; this is the standard spreading of these properties in a flat family [19, Tag 052A; Tag 0578; Tag 0559]. This proves the flatness and fibre assertions in (i), as well as the first assertion of (ii).

Since q is projective and the $\mathcal{A}_i$ are flat, their closed immersions define two sections of the relative Hilbert scheme $\mathrm{Hilb}_{\mathcal{I}/\mathcal{M}}$. Its diagonal is closed. The equalizer of the two sections is therefore closed and omits the generic point. Removing it makes $(\mathcal{A}_1)_x$ and $(\mathcal{A}_2)_x$ distinct for every geometric x.

Give $\mathcal{A} = \mathcal{A}_1 \cup \mathcal{A}_2$ its scheme-theoretic union structure. Let $\mathcal{J}$ be the ideal of $\mathcal{A} \cap \mathcal{I}^{\mathrm{ne}}$ in $\mathcal{I}^{\mathrm{ne}}$. The geometric generic fibre is reduced and consists exactly of the two generic components, so $\mathcal{J}$ vanishes there. The image of $\mathrm{Supp}(\mathcal{J})$ in the noetherian integral scheme $\mathcal{M}$ is constructible by Chevalley's theorem and does not contain its generic point. A dense constructible subset of an irreducible noetherian scheme contains the generic point, so this image is not dense and its closure is proper. Remove this closure. Then $\mathcal{J} = 0$, and

$$\mathcal{I}^{\mathrm{ne}} = \mathcal{A} \cap \mathcal{I}^{\mathrm{ne}}$$

over the remaining open. The fibrewise reducedness, the distinctness and geometric integrality of the two curve fibres, and this scheme-theoretic equality prove (ii) and (iii).

Denote the remaining open by $\mathcal{M}^\circ$. The morphism $\mathcal{M}^\circ \to \mathcal{B}$ is open because it is the restriction of the smooth morphism $\mathcal{M} \to \mathcal{B}$ [19, Tag 056G]. Hence its image $\mathcal{B}^\circ$ is nonempty and open. For $b \in \mathcal{B}^\circ(\mathbf{C})$, the fibre U_b is a nonempty open of the integral surface $(X_b)_{\mathrm{reg}}$; it is therefore dense, irreducible, and connected. The fibrewise conclusions (i)–(iii) give the final assertion. □

6. Pair convolution detects IC_X

Fix $b \in \mathcal{B}^\circ(\mathbf{C})$, put $(P, X) = (P_b, X_b)$, and let $U = U_b$ as in Theorem 5.1. Let $\rho : S \to X$ be the minimal resolution and let $j : S \to P$ be its composition with the inclusion. Put

$$K = \mathrm{IC}_X, \qquad \widetilde{K} = Rj_*\mathbf{Q}_S[2] = K \oplus \delta_0.$$

Consider

$$\widetilde{m} : S \times S \longrightarrow P, \qquad (p,q) \longmapsto j(p) + j(q),$$

and let $\sigma(p,q) = (q,p)$. A superscript $-$ denotes the anti-invariant direct summand under σ.

Lemma 6.1 (Top cohomology on the split-pair locus)**.** *After replacing U by a dense open subset, put $U^+ = -U$. For $t = -x \in U^+$, the fibre $F_t = \widetilde{m}^{-1}(t)$ has exactly four one-dimensional irreducible components in its reduction: the strict transforms of $(\mathcal{A}_1)_x$ and $(\mathcal{A}_2)_x$ and the two structural components*

$$E_{\mathrm{exc}} \times \{s_t\}, \qquad \{s_t\} \times E_{\mathrm{exc}}.$$

Here s_t is the unique point of S above t. The involution σ exchanges the first two components and exchanges the last two. Consequently

$$\dim \mathcal{H}^{-2}(\widetilde{K} * \widetilde{K})_{-,t} = 2, \qquad \dim \mathcal{H}^{-2}(K * K)_{-,t} = 1.$$

After a further shrinking of U, the restriction $\mathcal{H}^{-2}(K * K)_-|_{U^+}$ is the constant rank-one local system.

Proof. Fix $t \in U^+$, write $x = -t \in U$, and let $E_{\mathrm{exc}} = \rho^{-1}(0)$. By definition, C_x parametrizes ordered pairs whose sum is $-x = t$. Hence the fibre $F_t = \widetilde{m}^{-1}(t)$ has precisely four irreducible curve components: the strict transforms of $(\mathcal{A}_1)_x, (\mathcal{A}_2)_x$ and the two structural components

$$C_{1,t} = E_{\mathrm{exc}} \times \{s_t\}, \qquad C_{2,t} = \{s_t\} \times E_{\mathrm{exc}},$$

where s_t is the unique point above t. Indeed, away from E_{exc} the resolution is an isomorphism, while a pair with one entry on E_{exc} must have the other entry over t. Thus there are no further top-dimensional components. The involution σ exchanges $(\mathcal{A}_1)_x$ with $(\mathcal{A}_2)_x$ and $C_{1,t}$ with $C_{2,t}$.

The shifts are fixed by the identity

$$\widetilde{K} * \widetilde{K} \simeq R\widetilde{m}_* \mathbf{Q}_{S \times S}[4].$$

Proper base change therefore identifies

$$\mathcal{H}^{-2}(\widetilde{K} * \widetilde{K})_t = H^2(F_t, \mathbf{Q}).$$

The top cohomology of a proper complex curve is freely generated by the fundamental classes of the irreducible components of its reduction. Thus $H^2(F_t, \mathbf{Q})_-$ has the explicit basis

$$[\,(\mathcal{A}_1)_x\,] - [\,(\mathcal{A}_2)_x\,], \qquad [C_{1,t}] - [C_{2,t}].$$

Here and below the same notation is used for a curve and its strict transform in F_t. Holomorphic transposition preserves complex orientation, so

$$\dim \mathcal{H}^{-2}(\widetilde{K} * \widetilde{K})_{-,t} = 2. \tag{6.1}$$

Expand the decomposition $\widetilde{K} = K \oplus \delta_0$:

$$\widetilde{K} * \widetilde{K} = (K * K) \oplus (K * \delta_0) \oplus (\delta_0 * K) \oplus (\delta_0 * \delta_0).$$

At $t \neq 0$, the last term has zero stalk. The two middle terms are both canonically K, and the commutativity constraint exchanges them. Their anti-invariant part is therefore canonically one copy of K. Since t is a smooth point of the surface X, its

degree--2 stalk is the line generated by $[C_{1,t}] - [C_{2,t}]$. Taking dimensions in this direct-sum decomposition gives

$$\dim \mathcal{H}^{-2}(K * K)_{-,t} = 1. \tag{6.2}$$

There is no Koszul sign: K has geometric shift [2], and $(-1)^{2\cdot 2} = 1$.

Shrink U^+ once more and choose a topological stratification. The strict transforms of $\mathcal{A}_1$ and $\mathcal{A}_2$ are then flat over U^+; proper stratified local triviality makes all four relative fundamental classes locally constant. They are globally labelled as well. The first two come from $\mathcal{A}_1$ and $\mathcal{A}_2$ in Theorem 5.1, while the other two are

$$C_1 = E_{\mathrm{exc}} \times s, \qquad C_2 = s \times E_{\mathrm{exc}},$$

where $s : U^+ \to S$ is the unique lift of the inclusion $U^+ \subset X_{\mathrm{reg}}$. Thus monodromy cannot permute these four component classes. Since complex orientation fixes the sign of each fundamental class,

$$[\mathcal{A}_1] - [\mathcal{A}_2], \qquad [C_1] - [C_2]$$

form a global frame of the anti-invariant rank-two local system $\mathcal{H}^{-2}(\widetilde{K} * \widetilde{K})_-|_{U^+}$.

On U^+ the sheaf K is $\mathbf{Q}_{U^+}[2]$. Hence the anti-invariant part of $(K * \delta_0) \oplus (\delta_0 * K)$ is the constant line generated by $[C_1] - [C_2]$. The direct-sum decomposition above then identifies $\mathcal{H}^{-2}(K * K)_-|_{U^+}$ with the complementary constant line. This proves the last assertion. □

Lemma 6.2 (Vanishing away from the theta divisor)**.** *There is a dense open subset $W \subset P \setminus X$ such that, for every $t \in W$, the scheme $X \cap (t - X)$ is a smooth connected projective curve. The involution $p \mapsto t - p$ acts as the identity on its top cohomology. In particular,*

$$\mathcal{H}^{-2}(K * K)_-|_W = 0.$$

Proof. The sum map $m|_{X\times X} : X \times X \to P$ is surjective because two translates of the ample divisor X always meet. Moreover,

$$m\big((\{0\} \times X) \cup (X \times \{0\})\big) = X.$$

Hence, for $t \notin X$, the fibre lies entirely in $X_{\mathrm{reg}} \times X_{\mathrm{reg}}$. Generic smoothness for the restricted sum map therefore gives smoothness of the general fibre. Since $t \notin X$ and $\mathrm{Stab}_P(X) = \{0\}$, the translate $t - X$ does not contain X; its restriction to the integral Cohen–Macaulay surface X is therefore an effective ample Cartier divisor. Hartshorne's connectedness theorem, applied on X, shows that this divisor, namely $X \cap (t - X)$, is connected [8, Corollary III.7.9]. The involution $p \mapsto t - p$ preserves its complex orientation and hence acts by $+1$ on its one-dimensional top cohomology. Therefore

$$\mathcal{H}^{-2}(K * K)_- = 0 \quad \text{on a dense open subset of } P. \tag{6.3}$$

□

Lemma 6.3 (Extraction of the divisor-supported summand)**.** *In the decomposition-theorem splitting of $(K * K)_-$, the constant rank-one local system of Lemma 6.1 is the restriction of an unshifted IC_X summand. It cannot arise from a shifted simple summand with support P, or from a summand with any other support.*

Proof. The complex $(K*K)_-$ is a direct summand of the projective direct image of a pure polarizable Hodge module. By the decomposition theorem and semisimplicity [1, Théorème 6.2.5] and [18, Théorème 5.3.1], it is a direct sum of shifted simple intersection complexes. A summand with proper support containing the generic point of U^+ must have support X. Such a simple summand has the form $\mathrm{IC}_X(\mathcal{V})[r]$, where $\mathcal{V}$ is an irreducible local system on a dense smooth open $X_{\mathcal{V}} \subset X_{\mathrm{reg}}$. Its restriction to $U^+ \cap X_{\mathcal{V}}$ is $\mathcal{V}[2+r]$, so contribution to ordinary degree -2 forces $r = 0$.

A full-support summand cannot account for the line in (6.2). Only finitely many simple summands occur. Choose $a \in U^+$ in the two-dimensional stratum of a common Whitney stratification subordinate to X and to all these summands. Normal nonsingularity along this stratum identifies a sufficiently small stratified analytic neighbourhood with

$$(B^2 \times \Delta, B^2 \times \{0\}),$$

where $B^2 \subset X_{\mathrm{reg}}$ is a ball and Δ is a normal disk. Intermediate extension commutes with this external product. Hence a full-support simple perverse summand is locally

$$\mathcal{L}[2] \boxtimes j_{\Delta,!*}\mathcal{W}[1], \qquad j_\Delta : \Delta^* \hookrightarrow \Delta,$$

for local systems $\mathcal{L}$ and $\mathcal{W}$.

Write T for the local monodromy of $\mathcal{W}$. On a disk the middle extension has the explicit stalk

$$\mathcal{H}^k(i_0^* j_{\Delta,!*}\mathcal{W}[1]) = \begin{cases} \ker(T-1), & k = -1, \\ 0, & k \neq -1, \end{cases} \qquad i_0 : \{0\} \hookrightarrow \Delta.$$

Indeed $Rj_{\Delta,*}\mathcal{W}[1]$ has stalk cohomology $\ker(T-1)$ in degree -1 and $\mathrm{coker}(T-1)$ in degree 0; the middle extension removes exactly the quotient supported at the origin. It follows that an unshifted full-support perverse summand contributes along X only in ordinary degree -3. To contribute to ordinary degree -2 it must occur with shift $[-1]$. But on the dense smooth open of P the same shifted summand is a nonzero local system $\mathcal{E}[3][-1]$, hence also contributes to ordinary degree -2 there, contradicting (6.3). Thus no full-support summand contributes to the rank-one line on U^+.

The two relevant shifts are summarized by

full-support summand	general point of P	general point of X
$\mathrm{IC}_P(\mathcal{E})$	-3	-3
$\mathrm{IC}_P(\mathcal{E})[-1]$	-2	-2.

Supports of dimension at most one do not meet a generic point of U^+, and a different irreducible divisor cannot contain the nonempty open $U^+ \subset X$.

The local system $\mathcal{V}$ has rank one because its contribution to (6.2) has rank one. All preceding shrinkings may be made Zariski open. Thus $U^+ \cap X_{\mathcal{V}}$ is a nonempty Zariski-open subset of the smooth irreducible variety $X_{\mathcal{V}}$ and is connected. Its restriction there is the constant local system found in Lemma 6.1. The complement in $X_{\mathcal{V}}$ has complex codimension at least one. By general position, every loop in $X_{\mathcal{V}}$ is homotopic to one avoiding this complement, so inclusion induces a surjection

$$\pi_1(U^+ \cap X_{\mathcal{V}}) \twoheadrightarrow \pi_1(X_{\mathcal{V}}).$$

The monodromy representation of $\mathcal{V}$ is trivial on the source and hence on $\pi_1(X_{\mathcal{V}})$. Thus $\mathcal{V}$ is trivial and the summand is IC_X itself. □

Proposition 6.4 (Alternating support detector)**.** *In* $\mathrm{Perv}(P,\mathbf{Q})$*,* $K=\mathrm{IC}_X$ *is a direct summand of* ${}^pH^0((K*K)_-)$*. After tensoring with* $\mathbf{C}$*, its image in* $\overline{\mathrm{Perv}}(P)$ *is nonzero and is a direct summand of* $\mathrm{Alt}^2(K)$*.*

Proof. The first assertion is Lemma 6.3. Apply the quotient functor to $K\otimes_{\mathbf{Q}}\mathbf{C}$. It is perverse t-exact, convolution is t-exact in $\overline{D}(P)$, and all nonzero perverse-cohomological degrees of $K*K$ map to zero. Hence the image of ${}^pH^0((K*K)_-)$ is the anti-invariant summand of the tensor square of the image of K. Because the support dimension is even, this is $\mathrm{Alt}^2(K)$ under the fibre functor. The exact quotient functor preserves the direct summand supplied by Lemma 6.3; that summand is nonzero because $\chi(K)=7$. □

7. Identification of the full Tannaka group

In this section, fix $b\in\mathcal{B}^\circ(\mathbf{C})$, where $\mathcal{B}^\circ$ is the open obtained in Theorem 5.1. Let

$$G=G(K),\qquad V=\omega(K).$$

By (3.7), $\dim V=7$. The self-duality (3.16) gives a perfect evaluation pairing $K*K\to\delta_0$. The chosen symmetry is centred at the origin, so no translated skyscraper occurs, and $\dim X=2$ makes the pairing symmetric. Equivalently, the nondegenerate self-duality on the odd-dimensional space V cannot be alternating, so its bilinear form is necessarily symmetric. Therefore the full group embeds faithfully as

$$G\hookrightarrow \mathrm{O}(V). \tag{7.1}$$

The simple perverse sheaf K is nondivisible in the sense of [9, Section 3.3]: if $t_a^*K\simeq K$, then $X+a=X$, and (3.8) gives $a=0$. By [9, Proposition 3.3(2)],

$$V|_{G^0}\quad\text{is irreducible.} \tag{7.2}$$

This result does not require the ambient abelian variety to be simple.

Choose $\alpha\in\Omega$ as in Proposition 3.7. By Lemma 3.8, $H^0(-\otimes\mathcal{L}_\alpha)$ is a tensor fibre functor on the category generated by K. Since any two fibre functors over $\mathbf{C}$ are noncanonically isomorphic, it may be identified with ω. Let $G_P=G$, and let G_H be the Tannaka group of the pure Hodge module IC_X^H, acting faithfully on the same complex vector space V. By [12, Corollary 3.2], the Hodge decomposition is defined by a cocharacter $h:\mathbf{G}_m^2\to G_H$. The comparison theorem [12, Corollary 2.5] gives

$$G_P\lhd G_H,\qquad [G_P^0,G_P^0]=[G_H^0,G_H^0]=:G^*.$$

Since $G_P^0\subset G_H^0$, equation (7.2) implies that $V|_{G_H^0}$ is irreducible. Thus the connected centre Z_H^0 acts by scalar matrices. As G_H^0 is reductive, multiplication $Z_H^0\times G^*\to G_H^0$ is surjective with finite kernel.

Put $\mu(z)=h(z,z^{-1})$. The antidiagonal cocharacter must lie in the common derived group rather than merely in the Hodge group. This follows from the determinant. Since $\mathbf{G}_m$ is connected, $\mu(\mathbf{G}_m)\subset G_H^0$. By Proposition 3.7,

$$\det\mu(z)=z^{(-2)\cdot2+0\cdot3+2\cdot2}=1.$$

The determinant is trivial on the semisimple group G^* and therefore descends to a character

$$\overline{\det}:G_H^0/G^*\longrightarrow\mathbf{G}_m.$$

The quotient G_H^0/G^* is the image of Z_H^0. On Z_H^0 the determinant is $c \mapsto c^7$; since the faithful scalar action embeds Z_H^0 into $\mathbf{G}_m$, the induced character $\overline{\det}$ has finite kernel. Hence the image of the connected group $\mathbf{G}_m$ under μ in G_H^0/G^* is both connected and finite, and is therefore trivial. Consequently

$$\mu(\mathbf{G}_m) \subset G^*,$$

the common Hodge and perverse derived group. Its weights on V are

$$-2, 0, 2 \quad \text{with multiplicities } 2, 3, 2. \tag{7.3}$$

Applying the fibre functor to Proposition 6.4 gives

$$\mathrm{Hom}_G(V, \Lambda^2 V) \neq 0. \tag{7.4}$$

Proposition 7.1 (Connected Tannaka group)**.** *There is an isomorphism of pairs*

$$(G^0, V|_{G^0}) \simeq (G_2, V_7).$$

Proof. By (7.1) and (7.2), the connected centre of G^0 acts by orthogonal scalars and is therefore trivial. Thus G^0 is semisimple. An irreducible representation of a product of semisimple groups is an external tensor product. Since 7 is prime and the action is faithful, G^0 has one simple factor.

For a dominant weight λ, the Weyl dimension formula is

$$\dim V_\lambda = \prod_{\alpha>0} \frac{\langle \lambda + \rho, \alpha^\vee \rangle}{\langle \rho, \alpha^\vee \rangle}$$

(compare [6, Lecture 24]). It is strictly increasing when a nonzero dominant weight is added. Applying it to the fundamental weights of each simple type gives the following complete seven-dimensional list:

type	highest weight	dimension
A_1	$6\omega_1$	7
A_6	$\omega_1,\ \omega_6$	7
B_3	ω_1	7
G_2	ω_1	7.

The remaining types are excluded as follows. For A_r the fundamental dimensions are $\binom{r+1}{i}$. The only ranks below A_6 that could conceivably contribute dimension seven are A_2, A_3, A_4, A_5. The Weyl formula gives, respectively, the dimensions below the next jump as $3, 3, 6, 6$ (then 8), $4, 4, 6$ (then 10), $5, 5$ (then 10), and $6, 6$ (then 15). Thus none has a seven-dimensional simple module. For $r \geq 7$ even the standard A_r-module has dimension at least 8. For C_2 the fundamental dimensions are $4, 5$ and the next irreducible dimension is 10; for C_3 they are $6, 14, 14$ and $\dim V_{2\omega_1} = 21$; for C_r with $r \geq 4$ the standard module already has dimension at least 8. In type B_r, $r \geq 4$, the vector module has dimension at least 9, and in type D_r, $r \geq 4$, it has dimension at least 8; the spin modules do not produce dimension 7. Finally, the smallest nontrivial irreducible dimensions of F_4, E_6, E_7, E_8 are $26, 27, 56, 248$, whereas G_2 has its seven-dimensional fundamental module. This leaves exactly the four cases in the table.

The orthogonal self-duality of V excludes type A_6. In type A_1, the central element $-I \in \mathrm{SL}_2$ acts trivially on $\mathrm{Sym}^6(\mathbf{C}^2)$, so the faithful image is PGL_2; in type B_3, the seven-dimensional vector representation has faithful image SO_7. The remaining possibilities are therefore PGL_2 on $\mathrm{Sym}^6(\mathbf{C}^2)$, SO_7 on its standard module, and G_2 on V_7. For SO_7, the exterior square of the standard module is the

irreducible adjoint module, so $\mathrm{Hom}_{\mathrm{SO}_7}(V, \Lambda^2 V) = 0$, contrary to (7.4). For a non-trivial cocharacter of PGL_2, the weights on $\mathrm{Sym}^6(\mathbf{C}^2)$ are seven distinct numbers. With the primitive cocharacter of the adjoint torus (so that the usual SL_2 weights are divided by two), they are

$$-3, -2, -1, 0, 1, 2, 3,$$

and every other nonzero cocharacter multiplies them by a nonzero integer. The trivial cocharacter has one weight of multiplicity seven. Neither possibility agrees with (7.3). Hence $G^0 = G_2$. □

Corollary 7.2 (The full Tannaka group)**.** *There is an isomorphism of pairs*

$$(G, V) \simeq (G_2, V_7).$$

In particular, G is connected; this identifies the full Tannaka group, not only its identity component.

Proof. By Proposition 7.1, $G^0 \simeq G_2$ on V_7. Since $G^0 \lhd G \subset \mathrm{O}(V)$,

$$G \subset N_{\mathrm{O}(V)}(G_2) = G_2 \times \{\pm I\}.$$

Indeed, conjugation gives a homomorphism from the normalizer to $\mathrm{Aut}(G_2)$. Since $\mathrm{Out}(G_2) = 1$, multiplication by a suitable element of G_2 puts every normalizing element in the centralizer. Schur's lemma makes that centralizer scalar, and its intersection with $\mathrm{O}(V)$ is $\{\pm I\}$. If $G \neq G^0$, choose $g \in G \cap (-G_2)$ and write $g = (-I)h$ with $h \in G_2 = G^0$. Then $-I = gh^{-1} \in G$. But $-I$ acts by -1 on V and by $+1$ on $\Lambda^2 V$, contradicting (7.4). Hence $G = G^0 = G_2$. □

8. Specialization and an explicit example

Proposition 8.1 (Specialization of the intersection complex)**.** *For every $b \in \mathcal{B}(\mathbf{C})$ there is a smooth pointed curve through b, with generic point in $\mathcal{B}^\circ$, along which*

$$\psi(\mathrm{IC}_{X_{\bar{\eta}}}) \simeq \mathrm{IC}_{X_b}.$$

Consequently the specialization Tannaka group embeds as a closed subgroup of G_2, compatibly with the seven-dimensional tautological modules.

Proof. Choose a smooth curve $T \subset \mathcal{B}$ through b whose generic point lies in $\mathcal{B}^\circ$, and pass to the strict henselization at b. Let $\mathcal{N}_T$ be the relative norm fibre in $\mathcal{C}_T^{(3)}$, and let D_0 be the section consisting of the three ramification points. Set $\widehat{\mathcal{N}}_T = \mathrm{Bl}_{D_0} \mathcal{N}_T$.

This is a smooth proper family of surfaces. To check the assertion at the centre, label the branch points after an étale base change. An additive local coordinate on the elliptic curve and cube-root coordinates at the ramification points put the relative norm equation in the form

$$z_1^3 + z_2^3 + z_3^3 = 0$$

over the base. Indeed the three elliptic coordinates sum to zero at D_0, and the local ramification index is three at each point. The relative germ is therefore the product of the base with the cone over the Fermat plane cubic. Blowing up its vertex gives a smooth family and commutes with passage to fibres. Away from D_0 the norm fibres are smooth by Proposition 3.2; the relative norm fibre is flat, being an effective Cartier divisor in the smooth relative symmetric cube. Thus it is smooth there as well.

The Abel map gives a projective morphism $\widehat{j} : \widehat{\mathcal{N}}_T \to \mathcal{P}_T$. On each fibre it is birational onto X_t and an isomorphism away from zero. Its exceptional curves are the elliptic curve E_0 and the strict transform $\widetilde{\Gamma}$ of $|\kappa|$. As in Proposition 3.2, their intersection matrix is

$$\begin{pmatrix} -3 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \det = 2.$$

The semismall decomposition consequently gives, on both the geometric generic and closed fibres,

$$R\widehat{j}_{t,*}\mathbf{Q}_{\widehat{N}_t}[2] \simeq \mathrm{IC}_{X_t} \oplus \delta_0^{\oplus 2}. \tag{8.1}$$

The classes of the two relative exceptional curves supply constant skyscraper summands. Smooth base change for $\widehat{\mathcal{N}}_T \to T$ and proper base change for $\widehat{j}$ identify the nearby cycles of the left side with the corresponding closed-fibre direct image. Nearby cycles are perverse exact and send δ_0 to δ_0. Applying them to (8.1) and cancelling the two skyscraper summands proves the asserted isomorphism. Cancellation is valid because the closed-fibre direct image is semisimple, so the nearby intersection complex is a direct summand of a semisimple object.

For an abelian scheme in characteristic zero, nearby cycles preserve convolution, hypercohomology and character twists, and the specialization map on fundamental groups is an isomorphism. [13, Lemma 14.1 and Section 14] therefore gives

$$G(\mathrm{IC}_{X_b}) \hookrightarrow G(\mathrm{IC}_{X_{\bar{\eta}}}).$$

Here one may spread the curve and construction over a finitely generated subfield of $\mathbf{C}$, use characteristic-zero étale nearby cycles, and then use comparison with complex sheaves. Invariance under extension of algebraically closed fields [9, Corollary 4.4] identifies the generic group with the group computed in Corollary 7.2. The fibre functors in specialization identify their tautological modules. □

Theorem 8.2 (All allowed parameters)**.** *For every $b \in \mathcal{B}(\mathbf{C})$,*

$$\big(G(\mathrm{IC}_{X_b}), \omega(\mathrm{IC}_{X_b})\big) \simeq (G_2, V_7).$$

Proof. Write $H = G(\mathrm{IC}_{X_b})$. By Proposition 8.1, $H \subset G_2$ on V_7. The geometry and Hodge calculation of Section 3 apply to every $b \in \mathcal{B}$. In particular, the translation stabilizer is trivial, $V_7|_{H^0}$ is irreducible, and the antidiagonal Hodge cocharacter has weights $-2, 0, 2$ of multiplicities $2, 3, 2$ in $[H^0, H^0]$. The determinant argument preceding Proposition 7.1 uses only these properties and applies unchanged.

The orthogonal seven-dimensional list in the proof of Proposition 7.1 leaves PGL_2, SO_7 and G_2 as possibilities for H^0. The Hodge weights exclude PGL_2, and $\dim \mathrm{SO}_7 = 21 > 14 = \dim G_2$ excludes $\mathrm{SO}_7 \subset G_2$. Hence $H^0 = G_2$, and $G_2 = H^0 \subset H \subset G_2$ proves the assertion. □

Proof of Theorem 1.1. The relative construction in Section 4 gives a symmetric relatively ample effective Cartier divisor, flat with geometrically integral fibres. The polarization and resolution are those of Proposition 3.2; Corollary 3.4 gives $\chi(\mathrm{IC}_{X_b}) = 7$. The Tannaka pair is determined by Theorem 8.2. □

Corollary 8.3 (A rational example)**.** *The smooth cover*

$$E : y^2 = x^3 + 2, \qquad C : w^3 = y - 1$$

and its centred Prym theta surface, defined over $\mathbf{Q}$, realize (G_2, V_7) after base change to $\mathbf{C}$.

Proof. Here $\Delta_E = 108$, $\Delta_{\mathrm{br}} = 27$, and $c = 1$. Apply Theorem 8.2. □

Corollary 8.4 (Minimal ambient dimension for V_7)**.** *No intersection complex of an integral subvariety in an abelian variety of dimension at most two has the Tannaka pair* (G_2, V_7).

Proof. A point has Euler characteristic one. For an integral curve with normalization of genus g, its perverse intersection complex has Euler characteristic $2g-2$, which is even. A two-dimensional integral subvariety of an abelian surface is the whole surface, with Euler characteristic zero. None gives the required dimension seven. □

Appendix A. Exact algebraic computations

We record the exact calculations used in Section 4. The intrinsic argument of Lemma 4.5 and Proposition 4.6 proves the splitting of the factor-ordering cover without a coefficient expansion. The calculations below give the normalized conic coefficients, the finite-field irreducibility certificate for the residual octic, and one admissible specialization. Every finite-field reduction used in the proof is displayed.

A.1. **Normalized conic coefficients.** On the normalized chart $e_1 = v = 1$ the shifted curve is

$$Y^2+2cY = \lambda x^3+\big(1-\lambda(1-3e_2+3e_3)\big)x^2+\big(2c+\lambda(e_2^3-3e_2e_3+3e_3^2)\big)x-\lambda e_3^3. \tag{A.1}$$

For the section $s = us_0 + s_1 + zs_2$ one has

$$\alpha = u(x-Y)+(Y-e_3)-zY, \qquad \beta = e_2+z(x-e_3), \qquad \gamma = -1+e_2z.$$

Substitute these expressions into $\mathrm{Nm}(s) = \alpha^3+\beta^3Y+\gamma^3Y^2-3\alpha\beta\gamma Y$, reduce Y^2 using (A.1), and divide by the fixed line $Y-x$. The quotient $Q_s = q_{00}+q_{01}x+q_{02}Y+q_{11}x^2+q_{12}xY+q_{22}Y^2$ has

$$\begin{aligned} q_{00} &= \frac{2c+\lambda(e_2^3-3e_2e_3-3e_3^2u+3e_3^2)}{\lambda},\\ q_{01} &= \frac{1+\lambda(3e_2+3e_3u^2-3e_3-1)}{\lambda},\\ q_{11} &= -(u-1)(u^2+u+1), \end{aligned}$$

and

$$\begin{aligned} \lambda q_{02} &= 2cz^3+e_2^3\lambda z^3+3e_2^2\lambda uz-3e_2^2\lambda z-3e_2e_3\lambda uz^2-3e_2e_3\lambda z^3+3e_2e_3\lambda z^2\\ &\quad -3e_2\lambda u+3e_2\lambda-3e_3\lambda u^2-3e_3\lambda uz+6e_3\lambda u+3e_3\lambda z-3e_3\lambda-\lambda+1,\\ \lambda q_{12} &= 3e_2\lambda uz^2+3e_2\lambda z^3-3e_2\lambda z^2+2\lambda u^3+3\lambda u^2z-3\lambda u^2-3\lambda uz\\ &\quad -\lambda z^3+\lambda+z^3,\\ \lambda q_{22} &= -\lambda u^3-3\lambda u^2z+3\lambda u^2-3\lambda uz^2+6\lambda uz-3\lambda u\\ &\quad -\lambda z^3+3\lambda z^2-3\lambda z+\lambda+z^3. \end{aligned}$$

These six coefficients determine M_s and hence F_8. In particular, for

$$M_s = \begin{pmatrix} q_{00} & q_{01}/2 & q_{02}/2\\ q_{01}/2 & q_{11} & q_{12}/2\\ q_{02}/2 & q_{12}/2 & q_{22}\end{pmatrix}$$

the only determinant expansion used below is

$$4\det M_s = 4q_{00}q_{11}q_{22} - q_{00}q_{12}^2 - q_{11}q_{02}^2 - q_{22}q_{01}^2 + q_{01}q_{02}q_{12}. \tag{A.2}$$

For the simple-factor assertion in Proposition 4.2, specialize in characteristic zero to

$$(c, \lambda, e_1, e_2, e_3) = (1, 1, 1, 0, 2), \qquad (u, v, z) = (1 + t, 1, 1).$$

Then (A.1) becomes

$$Y^2 + 2Y = x^3 - 6x^2 + 14x - 8,$$

and

$$\alpha = (1 + t)(x - Y) - 2, \qquad \beta = x - 2, \qquad \gamma = -1.$$

The preceding formulas give

$$\begin{array}{lll} q_{00} = 2 - 12t, & q_{01} = 6t(t + 2), & q_{02} = -2(3t^2 + 3t - 1), \\ q_{11} = -t(t^2 + 3t + 3), & q_{12} = t(2t^2 + 6t + 3), & q_{22} = -t(t^2 + 3t + 3). \end{array}$$

The division by the fixed line is exact already in $\mathbf{Q}[t, x, Y]$:

$$\mathrm{Nm}(s) - (Y - x)Q_s = -(Y + 1)\big(Y^2 + 2Y - x^3 + 6x^2 - 14x + 8\big).$$

Substitution into (A.2) gives

$$4\det M_{1+t,1,1} = 2t(24t^3 + 56t^2 + 33t + 6). \tag{A.3}$$

In particular, $4F_8(1, 1, 1) = 12 \neq 0$ in $\mathbf{Q}$.

A.2. **Geometric integrality of the residual octic.** For this argument, specialize to characteristic 13 at $(c, \lambda, e_1, e_2, e_3) = (1, 1, 1, 0, 2)$ and put $H(u, v) = 4F_8(u, v, 1)$. To make the passage from the normalized formulas explicit, set

$$q_{ij}^{\mathrm{hom}}(u, v, z) = v^3 q_{ij}(u/v, 1, z/v), \qquad H(u, v) = \left.\frac{4\det M^{\mathrm{hom}}}{u - v}\right|_{z=1}.$$

Here M^{hom} is the symmetric matrix of (A.2) formed from the q_{ij}^{hom}. Expanding (A.2) gives

$$\begin{aligned} H(u, v) = {} & 6u^3v^5 - u^3v^4 + 6u^3v^3 - u^3v^2 - u^3v - 5u^2v^6 + 5u^2v^4 + 3u^2v^3 + 4u^2 \\ & + 5uv^7 + 3uv^6 - 3uv^4 + 3uv^3 + 4uv \\ & - 6v^8 - 2v^7 + 2v^6 + v^5 - 2v^4 + 4v^2. \end{aligned}$$

Its leading coefficient in v is $-6 = 7 \in \mathbf{F}_{13}^{\times}$, and

$$H(4, v) = 7h(v), \qquad h(v) = v^8 - 3v^7 - 2v^6 + 3v^5 + 4v^4 + 4v^3 - 3v^2 - 5v - 2.$$

Irreducibility follows from the standard factor-degree criterion. In $\mathbf{F}_{13}[v]/(h)$ all powers are reduced by the single relation

$$v^8 = 3v^7 + 2v^6 - 3v^5 - 4v^4 - 4v^3 + 3v^2 + 5v + 2. \tag{A.4}$$

Starting with $r_0 = v$, define r_{j+1} to be the degree-at-most-seven remainder of r_j^{13} upon division by h. Equivalently, $r_j \equiv v^{13^j} \pmod{h}$. Reduction by (A.4) gives

$$\begin{aligned}
r_0 &= v,\\
r_1 &= -v^7 + 2v^6 + 6v^5 - v^4 + 6v^3 - 2v^2 - 2v - 6,\\
r_2 &= -4v^7 + v^6 - 5v^5 - 3v^4 + 3v^3 - 3v^2 - 4v - 1,\\
r_3 &= -3v^7 - 3v^6 + 4v^5 + 6v^3 + 2v^2 + 5v - 2,\\
r_4 &= v^7 + 4v^6 + 2v^5 + 5v^4 - 4v^3 - 6v^2 - 2v + 4,\\
r_5 &= -2v^7 + 4v^6 + 3v^5 - 4v^4 - 3v^3 + 4v^2 - 3v - 4,\\
r_6 &= 2v^7 + 5v^6 - v^4 - 5v^3 + 4v^2 + 4,\\
r_7 &= -6v^7 + 3v^5 + 4v^4 - 3v^3 + v^2 + 5v - 5,\\
r_8 &= v.
\end{aligned}$$

Thus h divides $v^{13^8} - v$. Also $r_4 - v = r(v)$, where

$$r(v) = v^7 + 4v^6 + 2v^5 + 5v^4 - 4v^3 - 6v^2 - 3v + 4.$$

Moreover,

$$a(v)h(v) + b(v)r(v) = 1$$

with

$$\begin{aligned}
a(v) &= 5v^6 + 3v^5 - 3v^4 - v^3 - v^2 - 6v - 2,\\
b(v) &= -5v^7 + 6v^6 - 5v^5 + 3v^4 + 2v^3 - 4v^2 - 2v - 4.
\end{aligned}$$

Hence $\gcd(h, v^{13^4} - v) = 1$. Recall that $X^{q^n} - X$ is the product, without repetition, of the monic irreducible polynomials over $\mathbf{F}_q$ whose degrees divide n. Since $r_8 = v$, every irreducible factor of h has degree dividing 8; the displayed Bézout identity excludes degrees dividing 4. The only divisors of 8 are $1, 2, 4, 8$, so every irreducible factor of h has degree 8. Because $\deg h = 8$, the polynomial h is irreducible.

If H factored, the constant nonzero leading v-coefficient would force each positive-v-degree factor to retain its v-degree after $u = 4$, which would factor h; a v-degree-zero factor would be a unit. Thus H is irreducible. Its homogenization is the specialized form $4F_8$. Because H contains a nonzero v^8 term, z does not divide that homogenization; therefore any nontrivial homogeneous factorization would dehomogenize to a nontrivial factorization of H. The projective special fibre is thus irreducible. Finally

$$H(1, 6) = 0, \qquad (\partial_u H, \partial_v H)(1, 6) = (2, 0).$$

Thus it has a smooth $\mathbf{F}_{13}$-point. The field $\mathbf{F}_{13}$ is perfect, so this irreducible curve is geometrically reduced. If it were not geometrically irreducible, Frobenius would act transitively on its geometric components; a rational point would then lie on every component and would be singular. Hence the special octic is geometrically integral. This is the finite-field input in Lemma 4.3.

A.3. **An admissible specialization.** For Lemma 4.10 use

$$(c, \lambda, e_1, e_2, e_3) = (1, 1, 1, 0, 2), \qquad (u : v : z) = (3 : 5 : 1)$$

over $\mathbf{F}_{13}$. The shifted elliptic equation is

$$Y^2 + 2Y = x^3 - 6x^2 + x + 5,$$

and the branch cubic $Y = 0$ has discriminant 4. The fixed divisor is cut out by

$$p(w) = w^3 - w^2 - 2, \qquad \operatorname{disc}(p) = 1,$$

so it is reduced. On the normalized chart,

$$(U, Z) = (u/v, z/v) = (3/5, 1/5) = (11, 8) \quad \text{in } \mathbf{F}_{13}^2.$$

Consequently

$$\big(q_{ij}^{\hom}(3,5,1)\big)_{ij} = 5^3\big(q_{ij}(U,1,Z)\big)_{ij}\Big|_{(U,Z)=(11,8)} = (5,1,7,7,0,2),$$

where the entries are ordered as $q_{00}, q_{01}, q_{02}, q_{11}, q_{12}, q_{22}$. The residual conic therefore factors as

$$5 + x + 7Y + 7x^2 + 2Y^2 = 5(1 + 9x + Y)(1 + 12x + 3Y).$$

Thus the three norm lines are

$$L_0 : Y = x, \qquad L_1 : Y = 4x - 1, \qquad L_2 : Y = 4 - 4x.$$

Substitution into the elliptic equation gives the three line-section cubics

	cubic in x	discriminant
L_0	$x^3 + 6x^2 - x + 5$	4
L_1	$x^3 + 4x^2 + x + 6$	3
L_2	$x^3 + 4x^2 + 2x - 6$	5.

Hence all three intersections with E are transverse. To keep the sign convention fixed, set

$$\mathcal{R}(x, Y) = x^3 - 6x^2 + x + 5 - (Y^2 + 2Y),$$

so that $E = V(\mathcal{R})$. The residual lines meet the branch line at

$$L_1 \cap \{Y = 0\} = (10, 0), \qquad L_2 \cap \{Y = 0\} = (1, 0),$$

and

$$\mathcal{R}(10, 0) = 12, \qquad \mathcal{R}(1, 0) = 1.$$

Their pairwise intersections are

$$L_1 \cap L_2 = (12, 8), \qquad L_0 \cap L_1 = (9, 9), \qquad L_0 \cap L_2 = (6, 6),$$

with

$$\mathcal{R}(12, 8) = 8, \qquad \mathcal{R}(9, 9) = 2, \qquad \mathcal{R}(6, 6) = 2.$$

All five values are nonzero in $\mathbf{F}_{13}$. Thus the residual line sections avoid the branch divisor, and the three degree-three line sections are pairwise disjoint on E.

The two residual line sections are therefore finite *étale* away from the branch locus and occur with multiplicity one in the norm divisor. The local product formula for the norm then forces exactly one of the three sheets of $C \to E$ to carry a simple zero above each residual point, which is condition (iii) of Definition 4.7.

The canonical-model condition follows from the same functions used in Proposition 3.2. Because $\mathbf{F}_{13}$ contains ζ_3, the nine quadratic representatives split into

$$\langle 1, x, Y, x^2 \rangle, \qquad \langle w, xw, wY \rangle, \qquad \langle w^2, xw^2 \rangle.$$

Their pole orders at a point over O are $(0, 2, 3, 4)$, $(1, 3, 4)$, and $(2, 4)$, respectively. Thus the only canonical quadric is $X_0X_2 - X_1^2 = 0$, of rank three, and $|\kappa|$ is the unique g_3^1. A member of this pencil is cut out by $a + bw$, so $\operatorname{Nm}_\pi(a + bw) = a^3 + b^3Y$; in particular its norm line has zero x-coefficient. By contrast, L_0, L_1, L_2 all have nonzero x-coefficient. Their inverse degree-three divisors are therefore not linearly

equivalent to κ, hence lie in the $h^0 = 1$ Abel locus and map to the smooth theta locus. This verifies condition (iv) and completes the arithmetic specialization used in Lemma 4.10.

References

[1] A. A. Beilinson, J. Bernstein, and P. Deligne, *Faisceaux pervers*, Astérisque **100**, Société Mathématique de France, Paris, 1982.

[2] Christina Birkenhake and Herbert Lange, *Complex abelian varieties*, second ed., Grundlehren der mathematischen Wissenschaften, vol. 302, Springer-Verlag, Berlin, 2004.

[3] Jean-Paul Brasselet, Dũng Tráng Lê, and José Seade, *Euler obstruction and indices of vector fields*, Topology **39** (2000), no. 6, 1193–1208.

[4] Mark Andrea A. de Cataldo and Luca Migliorini, *The hard Lefschetz theorem and the topology of semismall maps*, Ann. Sci. Ècole Norm. Sup. (4) **35** (2002), no. 5, 759–772.

[5] J. Franecki and M. Kapranov, *The Gauss map and a noncompact Riemann–Roch formula for constructible sheaves on semiabelian varieties*, Duke Math. J. **104** (2000), no. 1, 171–180.

[6] William Fulton and Joe Harris, *Representation theory: A first course*, Graduate Texts in Mathematics, vol. 129, Springer-Verlag, New York, 1991.

[7] Alexander Grothendieck and Jean Dieudonné, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas, troisième partie*, Publ. Math. Inst. Hautes Études Sci. **28** (1966), 5–255.

[8] Robin Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. 52, Springer-Verlag, New York, 1977.

[9] Ariyan Javanpeykar, Thomas Krämer, Christian Lehn, and Marco Maculan, *The monodromy of families of subvarieties on abelian varieties*, Duke Math. J. **174** (2025), no. 6, 1045–1149,https://doi.org/10.1215/00127094-2024-0053.

[10] Masaki Kashiwara, *Index theorem for constructible sheaves*, in *Systèmes différentiels et singularités*, Astérisque **130** (1985), 193–209.

[11] George Kempf, *On the geometry of a theorem of Riemann*, Ann. of Math. (2) **98** (1973), no. 1, 178–185, https://doi.org/10.2307/1970910.

[12] Thomas Krämer, Christian Lehn, and Marco Maculan, *Exceptional Tannaka groups only arise from cubic threefolds*, J. Reine Angew. Math. **830** (2026), 51–99,https://doi.org/10.1515/crelle-2025-0073.

[13] Thomas Krämer and Rainer Weissauer, *Vanishing theorems for constructible sheaves on abelian varieties*, J. Algebraic Geom. **24** (2015), no. 3, 531–568, https://doi.org/10.1090/jag/645.

[14] Henry B. Laufer, *On minimally elliptic singularities*, Amer. J. Math. **99** (1977), no. 6, 1257–1295.

[15] Daniel Litt, *Problem no. 13*, Problems I Like, https://www.problemsilike.com/13, accessed September 30, 2026.

[16] I. G. Macdonald, *Symmetric products of an algebraic curve*, Topology **1** (1962), 319–343, https://doi.org/10.1016/0040-9383(62)90019-8.

[17] Henry C. Pinkham, *Simple elliptic singularities, del Pezzo surfaces and Cremona transformations*, in *Several complex variables*, Proc. Sympos. Pure Math., vol. 30, Part 1, American Mathematical Society, Providence, RI, 1977, pp. 69–71.

[18] Morihiko Saito, *Modules de Hodge polarisables*, Publ. Res. Inst. Math. Sci. **24** (1988), no. 6, 849–995, https://doi.org/10.2977/PRIMS/1195173930.

[19] The Stacks Project Authors, *Stacks Project*, https://stacks.math.columbia.edu, accessed August 11, 2026.